



2026 IEEE 8TH INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE CIRCUITS AND SYSTEMS

16-18 September, 2026
Ha Long, Vietnam

AICAS 2026

PROGRAM BOOKLET



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Message from General Chairs

On behalf of the Organizing Committee, we are pleased to welcome you to the 2026 IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS 2026), to be held from September 16 to 18, 2026, in the beautiful coastal city of Ha Long Bay, Vietnam. As the 8th edition of AICAS, this conference continues the IEEE Circuits and Systems Society (CASS) mission to promote innovation and collaboration at the frontier of artificial intelligence circuits and systems. Under the theme **"AI Circuits and Systems for a Sustainable Future,"** AICAS 2026 aims to bring together researchers, engineers, industry leaders, and young professionals worldwide to explore transformative technologies that enable the next generation of intelligent and sustainable computing systems.

Artificial intelligence is rapidly reshaping almost every aspect of modern society, creating unprecedented opportunities while also introducing new challenges in computing efficiency, scalability, and sustainability. The continuous evolution of AI applications, from edge intelligence and autonomous systems to large-scale cloud computing, demands innovative solutions across all layers of the computing stack, including algorithms, architectures, circuits, devices, and systems. AICAS serves as a unique international forum that bridges these disciplines, fostering the development of energy-efficient, intelligent, and reliable AI-enabled circuits and systems for future applications.

AICAS 2026 represents an important milestone for the global circuits and systems community. This is the first time AICAS has been hosted in Vietnam by Vietnam National University, Hanoi (VNU), and marks one of the largest IEEE CASS flagship conferences held in the country. Furthermore, AICAS 2026 marks the first joint organization of an IEEE AICAS flagship conference by IEEE CASS communities from both **China and Vietnam**, demonstrating strong international collaboration, a shared vision, and growing research excellence within the Asia-Pacific region. Through this collaboration, we hope to strengthen further connections among academia, industry, and professional communities, creating new opportunities for innovation and partnership in artificial intelligence circuits and systems.

We are proud to present an exciting, comprehensive technical program featuring four distinguished keynote talks by internationally recognized experts from academia and industry. These keynote presentations will provide valuable insights into emerging challenges and opportunities in AI hardware, intelligent computing, and future circuit and system technologies. In addition to the main technical sessions, AICAS 2026 will feature dedicated workshops and community activities designed to encourage deeper discussions and knowledge exchange. The conference will host the **S2CAS Workshop: Smart Sensor Circuit and Intelligent Systems for Emerging Applications**, focusing on emerging intelligent sensing technologies and their applications, and the Chiplet Workshop, which will address next-generation computing architectures and advanced system integration technologies.

Beyond the main conference program, AICAS 2026 introduces several initiatives to enrich participants' experience. Before the conference, a Seasonal School will be held in **Hanoi**, providing an educational platform for students, young researchers, and professionals to learn from leading experts in artificial intelligence circuits and systems. This initiative also offers participants the opportunity to experience Hanoi's rich cultural heritage before traveling to Ha Long Bay for the main conference activities. We believe these educational and cultural exchanges are essential to cultivating the next generation of researchers and strengthening the global AICAS community.

AICAS 2026 also strongly emphasizes community engagement and networking. Participants will have opportunities to connect with colleagues and establish new collaborations through a series of social activities. The Welcome Reception will be hosted in a unique cave setting in Ha Long Bay, allowing participants to experience the remarkable natural landscape of this world-renowned destination while interacting with members of the global CAS community. The Gala Banquet will be held as a special cruise experience on Ha Long Bay, providing an unforgettable evening to celebrate the conference's achievements and strengthen friendships among participants. In addition, the **Women in Circuits and Systems (WiCAS) and Young Professionals (YP) Plenary Talk** will provide an important platform to support diversity, inclusion, and the development of future leaders in the circuits and systems community.

The success of AICAS 2026 would not be possible without the dedication and contributions of many individuals and organizations. We sincerely appreciate the Technical Program Chairs, organizing committee members, reviewers, session chairs, volunteers, and all contributors whose commitment and hard work made this conference possible. We also extend our deepest gratitude to our industry partners and sponsors, including **T-Head, Cadence, and Huawei**, whose strong support highlights the close collaboration between academia and industry in advancing artificial intelligence circuits and systems.

Finally, we warmly welcome you to Ha Long Bay, Vietnam. Beyond the exciting technical discussions and professional interactions, we hope you will enjoy the host country's unique natural beauty, cultural richness, and hospitality. We sincerely hope that AICAS 2026 will provide an inspiring platform for exchanging ideas, building collaborations, and creating lasting memories within the global AI circuits and systems community.

We look forward to welcoming you to AICAS 2026 in Ha Long Bay and wish you a productive, inspiring, and memorable conference experience.

Prof. Yongfu Li
Prof. Xuan-Tu Tran
General Chairs, IEEE AICAS 2026

Message from Technical Program Committee Chairs

Artificial intelligence is transforming our world at an unprecedented pace. The remarkable progress of generative AI and Large Language Models (LLMs) has captured worldwide attention and is reshaping how we work, communicate, create, and interact with technology. Yet, AI is much more than LLMs. Realizing its full potential requires innovations across the entire computing stack—from devices and circuits to architectures, algorithms, and intelligent systems. AI Circuits and Systems play a vital role, encompassing AI hardware and accelerators, edge AI, neuromorphic and in-memory computing, intelligent sensing, energy-efficient architectures, AI-assisted design, and many other emerging technologies. Against this exciting backdrop, on behalf of the Technical Program Committee, we are pleased to welcome you to the 8th IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS 2026), held from September 16–18, 2026, in Ha Long, Vietnam.

AICAS 2026 has attracted strong interest from the international research community, receiving 330 paper submissions. After a rigorous peer-review process, 131 papers from authors in 35 countries and regions were selected for presentation at the conference and publication in the IEEE conference proceedings, for an acceptance rate of 39.7%. These contributions reflect the breadth and diversity of AICAS research and demonstrate the increasingly important role of circuits and systems in enabling the next generation of artificial intelligence.

The AICAS 2026 technical program includes 26 technical sessions, covering a diverse range of topics including AI for circuits and systems, circuits and systems for AI, AI computing architectures, computer vision, hardware/software co-design, in-memory computing, neuromorphic and bio-inspired systems, emerging devices, edge intelligence, AI-powered electronic design automation, wireless communications, and emerging AI applications. The program also features special sessions and the IEEE AICAS Grand Challenge, highlighting emerging research directions such as edge LLM acceleration, energy-efficient AI hardware, compute-in-memory co-design, and AI-assisted IC design.

We are also honored to feature four distinguished keynote speakers, addressing topics ranging from AI super-agents for autonomous chip design and neuromorphic intelligent vision to adaptive dataflow computing and infrastructure for the next AI era. In addition, the broader AICAS 2026 program includes two tutorial sessions on AI-enabled IC design and high-speed interconnects for AI computing infrastructures, as well as two workshops and an industry talk, further enriching opportunities for learning, discussion, and collaboration.

The success of the AICAS 2026 technical program would not have been possible without our community's dedication and hard work. We sincerely thank the Track Chairs, Special Session Chairs, TPC members, and reviewers for their tremendous efforts in managing the review process and ensuring the conference's high technical quality. We also thank the General Chairs, the Organizing Committee, the IEEE Circuits and Systems Society, the IEEE CAS Vietnam Chapter, the IEEE CAS Shanghai Chapters, and all supporting organizations.

Most importantly, we sincerely thank all the authors, speakers, and participants for choosing AICAS 2026 as a venue to share your research, ideas, and experiences. Your contributions are at the heart of the conference and help advance the rapidly evolving field of artificial intelligence circuits and systems.

We hope AICAS 2026 will offer an inspiring technical program, stimulating discussions, valuable connections, and new opportunities for collaboration. We also hope that you will enjoy Ha Long Bay, a UNESCO World Heritage Site renowned for its spectacular natural beauty and unique landscape.

We warmly welcome you to AICAS 2026 and wish you a productive, enjoyable, and memorable conference in Vietnam!

Li Du, Nanjing University, China

Duy-Hieu Bui, Vietnam National University, Hanoi

Xiaoqing Wen, Kyutech, Japan

Jun Zhou, University of Electronic Science and Technology of China

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Keynote Speakers



Ziyad Hanna
*Corporate VP, System
Design Verification and
AI Technologies at
Cadence Design Systems*

Keynote #1: From Design Automation to Autonomous Design: AI Super Agents for Chip Development

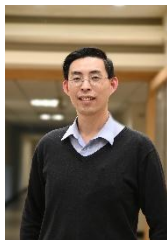
(9:30 - 10:30, September 16th)

Venue: Diamond 1, Wyndham Legend Halong

Chairs: Du Li, Nanjing University, China

Abstract: AI is already improving productivity in many tasks in chip design and verification, but today's solutions remain largely assistant-driven and task-specific. The next step is to evolve toward agentic AI systems that can plan, execute, evaluate, and refine complex design workflows across multiple tools and engineering domains. This talk examines how Super Agents, coordinated teams of specialized AI agents, can accelerate chip design and validation through iterative, closed-loop execution tightly integrated with EDA tools, engines, and design data. It will discuss potential architectures, including agent orchestration, multi-agent collaboration, EDA harnesses, and human-in-the-loop control, as well as the practical challenges of reliability, verification, security, data governance, and deployment at scale.

Biography: Ziyad Hanna is a distinguished global technology executive, computer scientist, and AI leader, with more than 35 years of experience shaping the fields of electronic design automation and artificial intelligence. As a Corporate Vice President of Cadence Design Systems and R&D General Manager, leading the company's technology innovations and global business, with a focus on software and AI technologies for the worldwide chip design industry. In 2025, Ziyad Hanna was appointed to lead Cadence's global artificial intelligence strategy, underscoring the company's deep trust in his visionary leadership and technical excellence. As Cadence's strategic head of AI, Ziyad shapes and executes the company's AI direction—driving innovation across global teams, identifying transformative opportunities, and steering technological advancement in one of the most critical areas of the future chip design ecosystem. His leadership in this role is pivotal to Cadence's evolution as a front-runner in AI-driven design and verification technologies. Before Cadence, he was Sr. Vice President, Chief Scientist, and General Manager of Jasper, the first startup to exit and managed by an Arab Israeli General Manager. While being a US permanent resident and leading the company from Silicon Valley, Jasper earned the Red Herring Top 100 North America award in 2013 and was acquired by Cadence in June 2014. Before joining Jasper, Ziyad was Senior Principal Engineer and Group Leader in the Design and Technology division at Intel Haifa. He played a key role in developing computer-aided design (CAD) systems that were deployed across nearly all Intel microprocessor designs since the early 1990s. A senior IEEE member and a lecturer in the EECS department at the Technion Israel, Ziyad is very active in the Hi-Tech field and has participated in many international conferences and workshops. He has published more than 30 articles, earned 25 patents, and given hundreds of talks; he is a frequent keynote speaker and a respected advisor in various fields of computer science, and he chaired conferences, including the annual Jasper User Group (JUG), for 15 consecutive years. Ziyad received numerous awards, including Intel's highest achievement award twice and the Cadence move-the-needle award. He earned his bachelor's and master's degrees from Tel Aviv University and a PhD in computer science from Oxford University in England, where he serves as a visiting professor in the computer science department. Ziyad is very active in fostering Hi-Tech in the Arab community. Ziyad was ranked number 39 in The-Marker's 100 most influential people in Israel for 2019, topping the list in the Hi-Tech category. He also received an honorary doctorate from Rubin College, and his name was included in the list of the most influential people in Hi-Tech by both The-Marker and People and Computers magazines.



Kea-Tiong (Samuel) Tang
National Tsing Hua
University,
Taiwan

Keynote #2: Learning from Biology - Neuromorphic Techniques for an Intelligent Vision System: Using Obstacle Avoidance as an Example

(11:00 - 12:00, September 16th)

Venue: Diamond 1, Wyndham Legend Halong

Chairs: Xuan-Tu Tran, Vietnam National University, Hanoi

Abstract: Massive computation, data, and power consumption limit the deployment of artificial intelligence on edge devices. Key neuromorphic computing technologies offer solutions for edge computing and are essential for developing next-generation AI chips. In this talk, I will discuss neuromorphic techniques, including models, architectures, and sensing. These techniques will be demonstrated on a microdrone platform for obstacle avoidance.

Biography: Dr. Kea-Tiong (Samuel) Tang received the B.S. degree in electrical engineering from National Taiwan University, Taiwan, in 1996, and received the M.S. and Ph.D. degrees in electrical engineering from California Institute of Technology, Pasadena, CA, USA, in 1998 and 2001, respectively.

From 2001–2006, Dr. Tang was a Senior Electrical Engineer at Second Sight Medical Products, Inc., Sylmar, CA, USA. He designed a mixed-signal ASIC for the Argus II Retinal Prosthesis System, which became the first FDA-approved retinal prosthesis. Since 2006, he has been a member of the Electrical Engineering Faculty at National Tsing Hua University, Hsinchu, Taiwan, and is currently a Distinguished Professor. His research interests include bio-inspired learning chips, miniature electronic systems, and biomedical implantable prosthetic devices. He has actively collaborated with multidisciplinary researchers and has published more than 300 peer-reviewed journal and conference papers in these research areas. He is a recipient of numerous awards, including the Outstanding Young Scholar Award, Wu Ta-You Memorial Award, National Innovation Award, Outstanding Electrical Engineering Professor Award, and the Outstanding Research Award, NSTC.

Dr. Tang is a member of IEEE CASS, SSCS, EDS, and EMBS. He served as Past Chair of the IEEE Biomedical and Life Science Circuits Systems Technical Committee (BioCAS). He was the Editor-in-Chief of IEEE Transactions on Biomedical Circuits and Systems (TBioCAS) (2022–2023). He has served as a TPC member of ISSCC, IEDM, ASSCC, ISCAS, and BioCAS. He was IEEE CAS Chapter Chair of the Taipei Section (2017–2018). He served as Chair of the IEEE Taipei Section (2021–2022). He served as a Board of Governors (BoG) member of CAS Society, representing Region 10 (2020–2021). He was the Vice President – Regional Activities and Membership of IEEE CASS (2022–2025). He is the current Vice President – Conferences of IEEE CASS. He was elected for Administrative Committee (AdCom) member for 2026–2028. Dr. Tang is an IEEE Fellow.



Tulika Mitra
National University of
Singapore, Singapore

Keynote #3: From FLOPS to Intelligence per Joule: The Case for Adaptive Dataflow

(9:30 - 10:30, September 17th)

Venue: Diamond 1, Wyndham Legend Halong

Chairs: Duy-Hieu Bui, Vietnam National University, Hanoi

Abstract: For decades, computing efficiency has been evaluated in terms of FLOPS or FLOPS per watt. Emerging AI workloads are forcing a different question: how much useful intelligence can be extracted from each joule?

In large AI models, sparsity is increasingly necessary to make computation and memory tractable during inference. But exploiting sparsity is not automatically beneficial: the cost of discovering, routing, and orchestrating useful work can rival or exceed the computation itself. This challenge is amplified as low-precision formats reduce arithmetic cost, shifting bottlenecks toward data movement and control.

In this talk, I will present adaptive dataflow as a foundation for efficient AI systems. Dataflow computing naturally exposes dependencies, exploits locality, and coordinates parallel execution. Adaptivity extends this paradigm to sparse, dynamic, and token-driven workloads whose useful structure may emerge only at runtime. I will argue that the central design question is not simply how to make dataflow more flexible, but where to introduce adaptivity, where to preserve static structure, and what that division of labor implies for the systems we build next.

Biography: Tulika Mitra is Provost's Chair Professor of Computer Science, Dean of Computing, and Vice Provost (Special Projects) at the National University of Singapore. Her research focuses on hardware-software co-designed systems that deliver more intelligence with less energy, while remaining fast, predictable, and reliable. She works on accelerator architectures and compilers, power and thermal management of heterogeneous systems, and worst-case execution time analysis of real-time systems. Her work has received the ESWEEK Test-of-Time Award and multiple Best Paper Awards. She has served as Editor-in-Chief of ACM Transactions on Embedded Computing Systems, General Chair and Program Chair of ICCAD, General Chair of ESWEEK, and is a recipient of the ACM SIGDA Distinguished Service Award.



Hong-Viet Le
FPT Smart Cloud,
Vietnam

Keynote #4: From Intelligence to Infrastructure: Building the Global Ecosystem for the Next AI Era

(11:00 - 12:00, September 16th)

Venue: Diamond 1, Wyndham Legend Halong

Chairs: Du Li, Nanjing University, China

Abstract: AI is shifting from experimental models to mission-critical infrastructure powering real-world applications across industries. The race is no longer just about algorithms. It is about the compute, and increasingly the chips, that power them. This keynote shares FPT's perspective and real-world experience on the global AI landscape and the strategic choices behind building a full-stack AI ecosystem across computing capabilities, technology, and partnerships.

Biography: For more than 15 years at FPT, Mr. Le Hong Viet has played a crucial role in FPT's technology growth and contributed to its international expansion. He is currently the CEO of FPT Smart Cloud, a flagship technology subsidiary of FPT that provides a world-class Artificial Intelligence platform and Cloud Computing services. Having pinpointed AI and Cloud as FPT's strategic technologies, Mr. Viet and FPT experts are focusing on developing 03 core product ecosystem - FPT.AI, FPT Cloud, and FPT AI Factory- through three layers: robust GPU Cloud infrastructure, AI platforms for AI model training and fine-tuning, and ready-to-deploy applications for enterprises. The strategy aims to deliver comprehensive applications within the FPT ecosystem, practical product and service packages for the market, and valuable resources for the community. In addition to serving as CEO of FPT Smart Cloud, Mr. Viet is a respected advisor and expert in Artificial Intelligence, Big Data, and Digital Transformation. He is also a senior advisor at FPT Ventures - the start-up investment fund that specializes in Technology.

Tutorial Speakers

Venue: ULIS – Jonathan KS. Choi Cultural Center
Vietnam National University, Hanoi

Sub-Theme #1: AI-Enabled IC Design Automation and Intelligent Hardware Architectures

Venue: 1st Floor, Jonathan KS Choi Culture Center, VNU Hanoi



Ibrahim Elfadel
Khalifa University,
United Arab Emirates

Hardware-Assisted Verification for Next-Generation AI Hardware Accelerators

(9:15 -10:15, September 14th)

Abstract: This tutorial addresses the verification gap created by the rapid growth of AI silicon. GPUs, NPUs, domain-specific ASICs, and chiplet-based AI systems now combine massive parallel compute fabrics, deep memory hierarchies, specialized interconnects, and complex software stacks, making traditional RTL simulation insufficient for validating designs that may contain tens of billions of transistors. The tutorial positions hardware-assisted verification (HAV) as essential infrastructure for pre-silicon AI hardware development, enabling real workloads to run before tapeout and supporting faster validation, debugging, and optimization.

The tutorial begins with a system-level introduction to AI accelerator architectures, including GPUs, NPUs, tensor processing arrays, dataflows, HBM, SRAM, NoCs, and chiplet-based systems. It then examines AI-specific verification challenges, including design scale, workload dependence across CNNs, transformers, and LLMs, hardware–software co-design, and non-deterministic debugging behavior. We then introduce HAV fundamentals, contrasting emulation, FPGA prototyping, and hybrid flows while discussing observability, debug, and scalability. The tutorial also compares major industry platforms and workflows, including Siemens Véloce, Cadence Palladium/Protium, and Synopsys ZeBu/HAPS.

A key tutorial theme is hardware–software co-verification. One important aspect is to explore how AI workloads can be executed pre-silicon to validate compiler mappings, runtime scheduling, numerical correctness, and precision behavior. The tutorial also connects verification to performance and energy optimization through workload profiling, memory/compute/interconnect bottleneck analysis, TOPS/W optimization, and design space exploration. Emerging AI hardware technology trends include chiplet-based accelerators, sparse and dynamic AI workloads, edge AI, low-power systems, and AI-driven CAD and verification.

AI hardware verification technology has moved beyond traditional RTL-centered verification toward workload-aware HAV, full-stack AI hardware/software co-validation, and scalable methodologies for future ultra-large AI accelerators. This tutorial's key outcome is to equip CAD researchers, EDA developers, AI hardware architects, verification engineers, compiler/runtime developers, and graduate students with modern verification workflows, while motivating new research directions in verification CAD for AI hardware.

Biography: Dr. Ibrahim (Abe) M. Elfadel is a Professor of Computer and Information Engineering at Khalifa University in Abu Dhabi, UAE, and VP-Initiatives of IEEE CEDA. Before joining academia, he served as a Senior Scientist at IBM in Yorktown Heights, where he contributed to the research, development, and deployment of CAD tools and methodologies for IBM high-end microprocessors. From 2012 to 2019, he led three industrially funded Abu Dhabi research centers focused on IoT, 3D Integration, and MEMS.

His recognitions include six IBM Invention Achievement Awards, an IBM Outstanding Technical Achievement Award, an IBM Research Division Award, the IEEE TCAD D. O. Pederson Best Paper Award, the SRC Board of Directors Award, the 2022 IFIP Service Award,

the 2023 Khalifa University Service Excellence Award, and the 2026 IEEE CASS John Choma Education Award.

Dr. Elfadel was the General Co-chair of VLSI-SoC 2017, PIERS 2025, and BioCAS 2025, and the Technical Program Co-chair of VLSI-SoC 2023, AICAS 2023, and BioCAS 2024. He serves on the editorial boards of IEEE TCASAI and Elsevier's Integration. He chairs the IEEE CASS Technical Committee on Education and Outreach and serves as Chair-Elect of the IEEE CASS Technical Committee on Machine Learning. He received his PhD from MIT in 1993.



Xinfei Guo
Shanghai Jiao Tong
University, China

AI-Driven Shift-Left for Circuit and System Design

(10:30 -11:30, September 14th)

Abstract: The shift-left paradigm in chip design automation seeks to move physically aware analysis into earlier design stages to improve design quality, reduce iteration costs, and enable predictive digital twins. However, a major bottleneck remains: determining exactly where early prediction is both highly accurate and practically impactful. This tutorial introduces cutting-edge, AI-fused shift-left methodologies spanning critical stages of the EDA flow, from pre-RTL design to signoff, covering both digital and analog regimes. Attendees will explore real-world case studies and state-of-the-art frameworks, including physically aware timing models integrated into logic synthesis, macro connectivity-aware optimization for early routability, fast timing estimation tightly coupled with ECO processes for accelerated closure, and large language model (LLM)-assisted analog circuit sizing. Concluding with a deep dive into the practical opportunities and open challenges—such as model accuracy, cross-node generalization, and seamless EDA toolchain integration—this tutorial equips researchers and practitioners with actionable insights into how AI-driven prediction is actively reshaping modern circuit and system design automation workflows.

Biography: Xinfei Guo is an Associate Professor at Shanghai Jiao Tong University (SJTU), where he directs the Intelligent Circuits, Architectures, and Systems (iCAS) Laboratory. He holds a Ph.D. in Computer Engineering from the University of Virginia and an M.S. from the University of Florida. Before entering academia, he gained extensive industry experience at NVIDIA and IBM Research in the United States. Dr. Guo's research spans software–hardware co-design for efficient AI systems, AI-driven electronic design automation (EDA), and resilient computing architectures. A prolific researcher, he has co-authored over 70 publications and three books and has earned several best paper and presentation awards. He holds several roles within the IEEE community, currently serving as Associate Editor-in-Chief of IEEE Transactions on VLSI Systems (TVLSI) and Associate Editor for Integration, the VLSI Journal. He also serves as General Co-Chair of IEEE MCSoc 2026, Publication Chair of AICAS 2026, and TPC Co-Chair of ISICAS 2025, and frequently contributes technical programs to DAC, ICCAD, and ISCAS. Dr. Guo is a Senior Member of both IEEE and ACM, and has been honored as an IEEE Circuits and Systems Society (CASS) Distinguished Lecturer for 2026–2027.



Nuno Horta
University of Lisbon,
Portugal

AI for Analog IC Design Automation: Evolution and Open Challenges

(13:15 - 14:45, September 14th)

Abstract: This tutorial provides an overview of the evolution of analog integrated circuit (IC) design automation, with a particular focus on the growing role of Artificial Intelligence (AI) in Electronic Design Automation (EDA). The tutorial begins with a brief overview of the IC and EDA markets, highlighting the strategic importance of semiconductor design and the increasing need for advanced automation tools. It then presents a historical perspective on key milestones in IC design and EDA, showing how design methods and tools have evolved to address growing circuit complexity and productivity demands. Building on this context, the tutorial discusses the evolution of analog IC design automation, from early manual and knowledge-based approaches to optimization-driven methods and, more recently, AI-based EDA techniques. The tutorial also introduces current commercial solutions and industrial trends, illustrating how AI is being incorporated into modern EDA tools to support designers, improve productivity, and accelerate design exploration. Finally, it concludes with a summary of the main challenges and opportunities in AI for analog IC design automation, including data availability, robustness, integration into industrial flows, and the balance between automation and designer expertise. Overall, the tutorial aims to give participants a clear, accessible understanding of how analog IC design automation has evolved, where AI-based approaches stand today, and what challenges remain for broader adoption in research and industry.

Biography: Nuno Horta (S'89–M'97–SM'11) received the Licenciado, MSc, PhD, and Habilitation degrees in Electrical Engineering from Instituto Superior Técnico (IST), University of Lisbon, Portugal, in 1989, 1992, 1997, and 2014, respectively. In 1998, he joined the Department of Electrical and Computer Engineering at IST, where he is currently a Full Professor and has served as Department Chair since 2025. He is also a Senior Researcher at Instituto de Telecomunicações, where he heads the Integrated Circuits Group. He has supervised more than 100 postgraduate works, including MSc and PhD theses. He has authored or co-authored more than 200 publications, including books, book chapters, international journal papers, and conference papers. He has also participated, as a researcher or coordinator, in several national and European R&D projects. He was or is General Chair of AACD 2014, PRIME 2016, SMACD 2016, and SMACD 2023, and has served on the organizing and technical program committees of several international conferences, including IEEE ISCAS, IEEE LASCAS, DATE, NGCAS, and ESSCIRC. His research interests are mainly in Electronic Design Automation and Applied Computational Intelligence, with applications in analog and mixed-signal IC design, analog IC design automation, and computational finance.



Hong Chen
Tsinghua University,
China

Bio-inspired neuromorphic olfaction: From algorithms to chips

(14:15 -15:15, September 14th)

Abstract: Inspired by the mammalian olfactory system, electronic noses (E-noses)—which are designed for automatic gas identification and analysis—are widely used in fields such as food safety, public safety, and environmental monitoring. With advances in microelectronics, E-noses have been evolving toward miniaturization, intelligence, and portability. In addition to rapidly detecting target gases, portable E-noses must also identify a wider variety of gas types. As application scenarios become more complex, demand for higher sensitivity and accuracy in portable E-noses is increasing. Furthermore, to ensure longer standby time, portable E-noses must feature low power consumption and high energy efficiency. To meet the requirements, researchers face three major challenges: (1) In terms of algorithms, it is essential to overcome catastrophic forgetting and develop a brain-inspired algorithmic model capable of identifying all target gas types within complex gas mixtures. (2) In terms of architecture, to enhance the learning capability of the neuromorphic olfactory processor, a general-purpose reconfigurable neuromorphic olfactory processor architecture that supports on-chip learning mechanisms must be proposed to overcome sensor drift while improving adaptability and applicability in dynamic environments. (3) In terms of circuit design, for complex gas analysis tasks, a conflict arises between the computational cost of algorithms and the constraints of chip power consumption. This tutorial introduces methods to address these challenges—from biological mechanisms to SNN models and their hardware implementation—and demonstrates the design of an asynchronous neuromorphic olfactory processor that balances low power consumption with high efficiency through a software–hardware co-design approach tailored to olfactory algorithms.

Biography: Dr. Hong Chen received the Ph.D. degree from the Department of EE at Tsinghua University in 2005. Since 2007, she has been working with the School of Integrated Circuits at Tsinghua University and is now a full professor. Her research interests include bio-inspired asynchronous neuromorphic circuits, biomedical circuits and systems, and so on. She has published over 140 papers at ISSCC, VLSI, DAC, ISCA, A-SSCC, ICML, ICLR, JSSC, and so on, and holds 40 patents. Dr. Chen is currently the associate editor of IEEE Transactions on Biomedical Circuits and Systems, a TPC member of IEEE ASYNC and IEEE ISCAS, and the general chair of IEEE ASYNC 2023. Dr. Chen has won awards including the Best Demo Award at ISCAS 2013, the Best Paper Nominee at ASYNC2021, ASYNC2023, and ASYNC2026, the Best Paper at ICCBD+AI 2024, and the First and Second prizes of the Beijing and National Science and Technology Progress Award in 2020 and 2023, respectively.



Amir Zjajo
Innatera Nanosystems
B.V., The Netherlands

Product-Ready Neuromorphic MCU at the Sensor Edge

(15:30 -16:30, September 14th)

Abstract: A quest to minimize energy per inference requires i) co-design across the compute stack, enabling the algorithms and applications to influence the underlying hardware design, ii) a large degree of hardware-software co-optimization (e.g., towards minimizing network size for given accuracy, or maximizing the number of operations within the allowed power envelope), and iii) a fully modular and flexible hardware system to unify the design process. In this tutorial, we formulate a hierarchical, modular neuromorphic framework that enables product-ready hardware-software co-design and efficient deployment of signal-processing workloads on neuromorphic SoCs. We exploit hardware-software synergy to examine omnidirectional dependencies across the entire design stack to optimize and/or satisfy smart sensing design constraints such as energy efficiency, performance, cost, and time-to-market. In particular, we emphasize trade-offs between biological and artificial circuits, focusing on continuous-time processing efficiency: we examine electrochemically accurate, multi-compartment neurosynaptic computational elements, abstract their fundamental functions by extracting the underlying dynamics, and analyze their complexity, accuracy, and flexibility in signal processing for time-varying tasks.

Strategies to maximize energy-efficiency, latency of neuromorphic SoCs that incorporate spiking neural network (SNN) cores: we follow through on several design principles, among others, distributed computation, specialization of neural designs at different scales, wiring cost minimization, continuous-time processing to enable high information rates, compartmentalization and complexing/extending of synaptic capabilities for reliable and efficient (resource-aware) signal processing, etc.

Product-ready modular neuromorphic SoC design methodology and trade-offs: we define a full set of requirements for a product-ready system-on-chip at the sensor edge and emphasize the communication paradigm between software-driven blocks and hard IPs, programmability and controllability of hard IPs, and DfT requirements.

Insights into software tools that aid both design and deployment of SNN accelerators and their ease of use: we highlight software tooling required for efficient network design and application deployment on SoC architectures that incorporate SNN cores.

(Live demo) A quantitative validation of SNN accelerators' competitive performance across various applications: In the evaluation, we include both SNN and non-SNN approaches using general, task-level benchmarking and hierarchical metric definitions that capture key performance indicators of interest, and we use common open-source benchmark tools that facilitate actionable benchmark implementation.

Biography: Dr. Amir Zjajo is co-founder of Innatera Nanosystems B.V. and serves as its Chief Scientist. Previously, he was a member of the research staff at Philips Research Laboratories and in Corporate Research at NXP Semiconductors. He was at Delft University of Technology, where he led research into intelligent neuromorphic systems. Dr. Zjajo has published 3 books, 3 book chapters, and more than 100 papers in mixed-signal VLSI design and neuromorphic circuits and systems, and holds more than 20 US patents or patent-pending applications. He received the M.Sc. and DIC degrees from Imperial College London, U.K., and the PhD degree from Eindhoven University of Technology, The Netherlands. His research interests include energy-efficient circuit and system design for neuromorphic signal processing, and bionic electronic circuits for autonomous cognitive systems. Dr. Zjajo won best paper award at BioDevices'15, LifeTech'19, and AICAS'23. He is a senior member of IEEE.



Wei Mao
Xidian University, China

Reconfigurable Accelerator Design for Multi-Precision Floating-Point Computing: From Processing Elements to SME-Oriented Architectures

(16:30 -17:30, September 14th)

Abstract: Reconfigurable Accelerator Design for Multi-Precision Floating-Point Computing: From Processing Elements to SME-Oriented Architectures. The convergence of High-Performance Computing (HPC) and Artificial Intelligence (AI) demands diverse computing precisions, exposing the limitations of traditional architectures that suffer from physically isolated datapaths, low hardware utilization, and high energy overhead. To address these challenges, this tutorial first introduces a reconfigurable multi-precision computing unit (PE) and a high-throughput vector systolic array. By proposing a 12-bit redundancy-minimized bit-splitting method based on low-precision combination, the design achieves 100% multiplier utilization across FP64, FP32, and FP16 formats. To further optimize performance, we detail novel hardware structures, including a Cascaded Enumeration Comparison (CEC) strategy to drastically reduce critical path delay and a dual-output Carry Select Adder (CSLA) that significantly optimizes the power-delay product. The array-level design uses orthogonal output flows and efficient data reuse to minimize partial-sum accumulation latency and memory-bandwidth bottlenecks. The tutorial also explores a unified architecture that natively fuses floating-point and fixed-point calculations to adapt to the fragmented precision requirements of modern AI, including emerging formats like BF16 and TF32. Built upon a universal 4-bit unit, this architecture supports diverse training and inference scenarios within a single PE. We present a highly optimized six-stage pipeline that uses Pipelined Enumeration Comparison (PEC) to handle multi-input exponent comparisons. Finally, we demonstrate a specialized multi-precision accelerator microarchitecture tailored for the ARM Scalable Matrix Extension (SME) instruction set, detailing its reconfigurable outer- and inner-product array configurations, along with a comprehensive UVM platform.

Biography: Wei Mao (Senior Member, IEEE) received the B.Eng. degree from Southeast University (SEU) in 2011 and the Ph.D. degree from the National University of Singapore (NUS) in 2017. From 2017 to 2019, he worked as an Analog IC Designer at HiSilicon Technologies Co., Ltd. He then served as a Research Associate Professor at the Southern University of Science and Technology from 2019 to 2023. He is currently an Associate Professor at Xidian University, Xi'an, China. His research interests include emerging computing and memory chip design for AI applications. He has published over 80 papers in top-tier IEEE/ACM journals and conferences, as well as one first-author monograph and two book chapters. His awards include the 2020 Wu Wenjun AI Science and Technology Award and the 2021 IEEE CAS Society Regional Chapter of the Year Award. He serves as an Associate Editor for the IEEE ICAS and a Guest Editor for IEEE TCAS-I. He has also served on the organizing or technical program committees of several renowned conferences, such as ISCAS, ISICAS, AICAS, BioCAS, APCCAS, VLSI-SoC, and CCF DAC.



Tran Thuy Vy
Cadence Vietnam Site
Leader, Senior
Application Engineer
Manager

AI-Driven EDA: Reshaping Chip Design in the AI era

(17:30 -18:00, September 14th)

Abstract: The semiconductor industry is undergoing rapid AI-driven transformation, redefining how chips are designed and built. This talk examines that shift through Cadence's two strategic pillars: Design for AI — building purpose-built tools and agent-ready design flows to enable next-generation AI silicon; and AI for Design — embedding AI into the EDA process itself through Cadence AI Super Agents, where AI Agents automate the EDA flow itself, from digital implementation to formal verification, simulation, and debug. Building on this, the talk introduces Cadence's Intelligent System Design vision—the convergence of AI Agents across digital, custom design, verification, PCB, 3D-IC, and multiphysics. It highlights AI-driven EDA as the key trend addressing rising design complexity, shrinking time-to-tapeout, and growing performance demands across the electronics industry.

Biography: Tran Thuy Vy - Vietnam Site Leader and Senior Application Engineering Manager at Cadence Design Systems, with over 20 years of experience in IC design and EDA technologies. She has worked on advanced-process-node chip design across multiple engineering roles. Her interests include digital implementation, verification, design automation, and AI-driven chip design methodologies. She is passionate about advancing design methodologies and growing Vietnam's semiconductor ecosystem through industry-academia collaboration.

Sub-Theme #2: High-Speed Interconnect and Clock Circuits for AI Computing Infrastructures

Venue: 2nd Floor, Jonathan KS Choi Culture Center, VNU Hanoi



Quan Pan
Southern University of
Science and Technology,
China

Introduction to High-Speed SerDes Communications

(9:15 -10:15, September 14th)

Abstract: Low-power wireline integrated circuits have become extremely attractive since they are extensively adopted in high-speed communications, such as local area networks, board-to-board, and data center-to-data center. The book first covers basic concepts in high-speed SerDes, including transceiver architecture and fundamentals, modulation, and channels. The paper then studies energy-efficient high-speed links with sophisticated equalization, including analog front-ends and high-speed drivers, and finally discusses crosstalk cancellation in multi-channel systems.

Biography: Quan Pan (Senior Member, IEEE) received his B.S degree in Electrical Engineering (EE) at the University of Science and Technology of China (USTC) in 2005, and his Ph.D. degree in Electronics and Computer Engineering (ECE) at the Hong Kong University of Science and Technology (HKUST) in 2014. From 2014 to 2018, he was a Senior Staff Engineer at a Silicon Valley startup, working on 400GbE high-speed SerDes. He joined the School of Microelectronics at Southern University of Science and Technology as an assistant professor in 2018 and is now a tenured full professor. His research interests include High-speed optical transceiver, wireless and wireline circuit design. Dr. Pan has contributed more than 100 peer-reviewed articles. He received the 2017 Outstanding Young Author Award from the IEEE Circuits and Systems Society. He serves as an active reviewer for many international journals, including JSSC, TCAS, TVLSI, JLT, PTL, JoS, and *et al.*



Xiaoyan Gui
Xi'an Jiaotong University,
China

High-Speed Ultra-Low Latency Mixed-Signal SerDes Transceiver Design Techniques for AI Computing Infrastructures

(10:30 -11:30, September 14th)

Abstract: AI computing's explosive growth, driven by trillion-parameter models, has rendered high-speed interconnections the critical data backbone for modern infrastructures, demanding designs that overcome AI-specific circuit challenges: the need for high-speed ultra-low latency SerDes transceivers to meet ns-level requirements for real-time inference and training. This tutorial provides a focused study on advanced high-speed ultra-low latency mixed-signal SerDes transceiver design techniques tailored to these demands, detailing the implementation of 60Gb/s, 112-Gb/s, and up to 200-Gb/s mixed-signal SerDes interfaces (via topology and CDR optimization), thereby equipping researchers, engineers, and graduate students with the systematic knowledge needed to develop high-performance interconnection circuits for next-generation AI systems.

Biography: Xiaoyan Gui (Senior Member, IEEE) received his Ph.D. degree from the University of California, Irvine, US., in 2011. He worked at Broadcom in Irvine from 2008 to 2012, where he worked on high-speed SerDes transceiver design as a design engineer and staff scientist. In 2012, he joined the faculty in the School of Information and Electronics at Beijing Institute of Technology (BIT), China, and later, in 2017, the faculty in the School of Microelectronics at Xi'an Jiaotong University (XJTU), China, where he is currently a Full Professor. His research covers CMOS high-speed wireline and wireless communications integrated circuit design. He is the co-author of more than 70 peer-reviewed journal and conference papers, including ISSCC and JSSC. He received the 2007 Henry Samueli Fellowship from the UCI School of Engineering and the 2022 National Teaching Achievement Award in Higher Education from

the Ministry of Education of P. R. China. Dr. Gui serves as TPC co-chair (2022) and TPC chair (2023) of IEEE ICTA.



Hao Xu
Fudan University, China

Clock Generation Techniques for High-Speed Wireline Transceivers

(13:15 -14:15, September 14th)

Abstract: High-speed wireline transceivers are critical circuit building blocks enabling the large-scale deployment of scale-up and scale-out data center infrastructures. Generating high-frequency, low-jitter clocks has become increasingly challenging for 200G/400G data links. This presentation introduces circuit techniques in frequency synthesis, clock distribution, and multi-phase clock generation for high-speed interfaces. We will explain the key evolution in detail.

Biography: Hao Xu received the B.S. degree in microelectronics from Fudan University, Shanghai, China, in 2010, and the M.S. and Ph.D. degrees in electrical engineering from the University of California, Los Angeles, in 2012 and 2018, respectively. Between 2017 and 2019, he worked at Broadcom Inc., Irvine, CA, USA, developing ADC-based high-speed optical transceivers. From 2019 to 2021, he worked at Apple Inc. in San Diego, CA, USA, developing high-performance cellular RF transceivers. He held various part-time positions with Broadcom, MediaTek, and Tensorcom during his graduate studies. Since 2021, he has been a faculty member at Fudan University. His research interests include mixed-signal, RF, and analog integrated circuit design.



Xiaoteng Zhao
Xidian University, China

Clock and Data Synchronization Techniques for High-Performance Interconnection

(14:15 -15:15, September 14th)

Abstract: This tutorial covers clock and data synchronization techniques for high-performance interconnection, including serial and parallel interface architectures. For serial interconnections, we review clock and data recovery (CDR) architectures including phase-locked loop (PLL)-based CDR, phase-interpolation (PI)-based CDR, baud-rate CDR, burst-mode CDR, and reference-less CDR, et., al. For parallel interconnections, we emphasize deskew technologies in forward clock (FC) architectures, which are critical for mitigating skew-induced timing errors in wide-parallel, high-throughput interconnections. This tutorial aims to provide designers with insights to optimize synchronization for next-generation high-performance interconnections.

Biography: Xiaoteng Zhao is a Professor at Xidian University. His research focuses on high-speed serial interfaces and chiplet interconnection integrated circuit design. He presides over one Key Program of the Major Research Plan and one General Program funded by the National Natural Science Foundation of China. He has authored over 50 peer-reviewed papers published in top-tier IEEE journals and conferences, including ISSCC, JSSC, CICC, TMTT, ESSERC, ASSCC, RFIC Symposium, and TCAS-I. One of his representative works was shortlisted for the Top 10 Semiconductor Research Advances in China (2022). He holds more than 30 authorized patents and software copyrights, and has received three technical innovation awards and best paper honors. He participated in formulating two IEEE industry standards and has served on the Technical Program Committee (TPC) of IEEE ICTA for three consecutive sessions.



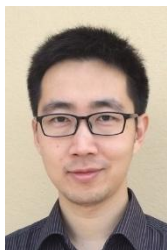
Zhao Zhang
Institute of
Semiconductors, Chinese
Academy of Sciences,
China

Design of Power-Efficient Reference-less PAM₄ CDR for Ultra-High-Speed Wireline Communication

(15:30 -16:30, September 14th)

Abstract: Referenceless CDR is flexible and a low-cost and low-power choice for wireline transceivers because it is free of an external reference clock and the global clock distribution network in multi-lane RX. Although referenceless CDR is popular for NRZ applications with data rates below 32Gbps, it remains challenging for PAM₄ design, preventing its use in over-100Gb/s wireline communication. In this tutorial, we first introduce the basics of NRZ reference-less CDR, then show how to migrate an NRZ design to its PAM₄ counterpart. Finally, we briefly review three design examples from the presenters' recent research, including a 64-Gb/s design, a 6-64-Gb/s design, and a 112-Gb/s design.

Biography: Zhao Zhang is now a Full Professor at the Institute of Semiconductors, Chinese Academy of Sciences, Beijing. His research interests include wireline transceiver design, low-jitter and low-power PLLs, and ultra-low-voltage ICs for bio/energy-harvesting applications.



Dan Li
Xi'an Jiaotong University,
China

Optical Receiver Front-Ends for AI Interconnect: Scenarios, Tradeoffs, and Design Guidelines

(16:30 -17:30, September 14th)

Abstract: The rapid growth of large AI models is driving increasingly distributed computing platforms, making high-bandwidth, energy-efficient communication between compute nodes more critical than ever. As interconnects become a key system bottleneck, the limitations of electrical links in bandwidth density, reach, and power make optical interconnects increasingly attractive. Meanwhile, optical links are evolving across multiple deployment forms, including pluggable optics, LPO, co-packaged optics (CPO), and optical I/O (OIO). These deployment forms do not impose the same receiver requirements: module-based and LPO links, switch-side or package-proximal CPO links, and compute-proximal OIO links can lead to substantially different design priorities for optical receiver front-ends. This tutorial presents a scenario-driven view of optical receiver front-ends for AI interconnects. The focus is on the receiver side, emphasizing the front-end tradeoffs that most strongly affect practical designs, including sensitivity, noise, bandwidth, tolerance to input parasitics, dynamic range, power efficiency, and technology choice, especially CMOS versus SiGe. By comparing representative AI optical interconnect scenarios, the tutorial will show how receiver requirements shift with system context and integration level. Attendees will gain a clearer picture of optical receiver front-ends for AI interconnects, along with practical design guidelines and a forward-looking view of future research and design opportunities.

Biography: Dan Li received the B.E. and M.E. degrees from Northwestern Polytechnical University, Xi'an, China, in 2004 and 2007, respectively, and the Ph.D. degree from the University of Pavia, Italy, in 2013. From 2007 to 2009, he worked at NVIDIA Shanghai R&D Center on custom SRAM circuitry. From 2011 to 2014, he worked at STMicroelectronics in Pavia, Italy, on CMOS optical receivers for 100GbE optical links and silicon photonics applications. He joined Xi'an Jiaotong University in 2015 and is currently a professor. His research interests include high-speed optical receivers, low-noise transimpedance amplifiers, 3D sensing, and low-power mixed-signal circuits. He has published in IEEE JSSC, TCAS-I, TCAS-II, JLT, PTL, and ISCAS, and has delivered invited talks and tutorials on low-noise optical receiver front-end design. He currently serves as an Associate Editor of IEEE Transactions on Circuits and Systems II: Express Briefs, and has also served the community in technical and organizing roles for IEEE conferences including ICTA, BioCAS, and ICECS.

Satellite Workshop

Venue: Wyndham Legend Ha Long Hotel, Ha Long Bay, Vietnam

S2CAS: Smart Sensor Circuit and Intelligent Systems for Emerging Applications

Venue: Panorama 2, Wyndham Legend Ha Long Hotel, Ha Long Bay, Vietnam

Chairs: Qing Zhang, Shenzhen University, China; Milin Zhang, Tsinghua University, China

Abstract: As the rapid proliferation of smart sensors, edge AI, and Internet-of-Things (IoT) systems drives the next wave of electronic innovation, the S2CAS Workshop (Smart Sensor Circuits and Intelligent Systems for Emerging Applications) addresses the critical co-design of sensing hardware, intelligent circuits, and application-driven systems under the “Sensor + X” paradigm - covering AI, healthcare, robotics, novel materials, and environmental monitoring. This workshop uniquely bridges device materials, analog/mixed-signal circuits, AI accelerators, and real-world deployment, positioning the IEEE CASS community as the central hub for end-to-end intelligent sensing solutions.

Organizing Committee

Milin Zhang, *Tsinghua University, China*

Xiaojin Zhao, *Shenzhen University, China*

Amine Bermak, *Hamad Bin Khalifa University, Qatar*

Wei Tang, *New Mexico State University, USA*

Qing Zhang, *Shenzhen University, China*

Invited Speakers

Hong Chen, *Tsinghua University, China*

Yi (Estelle) Wang, *AUMOVIO, Singapore*

Jianbiao Xiao, *Chip-Artisan Tech. Corp., China*

Weiwei Shi, *Shenzhen University, China*

The 2nd IEEE Chiplet Workshop on Circuits and Systems for Next-Generation AI Computing

Venue: Panorama 1, Wyndham Legend Ha Long Hotel, Ha Long Bay, Vietnam

Chairs: Chixiao Chen, Fudan University, China; Anh Tuan Do, A*STAR, Singapore.

Workshop Overview

The 2nd IEEE Chiplet Workshop on Circuits and Systems for Next-Generation AI Computing (CSNAIC) is a chiplet event of AICAS 2026, sponsored by the IEEE Circuits and Systems Society (CASS). This year features a dual-venue format: Ha Long Bay, Vietnam, on 18 September 2026 (co-located with AICAS 2026), and Shanghai, China, on 9 October 2026.

The workshop brings together researchers and industry experts to discuss emerging challenges and innovative solutions in chiplet-based architectures, advanced circuits and systems, and EDA technologies for next-generation AI computing.

Topics of Interest

- Chiplet architectures and 2.5D/3D integration
- High-speed interconnects and advanced packaging
- Compute-in-memory and near-memory computing
- Heterogeneous AI accelerator design
- Reconfigurable and low-power edge AI systems
- EDA methodologies for chiplet-based systems
- Standardization for scalable chiplet ecosystems

Workshop Highlights

- Five invited talks by leading international researchers
- Dual-venue format: Vietnam (AICAS) + Shanghai
- Poster Session & Call for Papers in Shanghai
- Sponsored by IEEE Circuits and Systems Society
- Networking with academia and industry experts

Organizing Committee

Chixiao Chen, *Workshop Co-Chair, Fudan University, China*

Anh Tuan Do, *Workshop Co-Chair, A*STAR, Singapore*

Program at a Glance

Venue: Wyndham Legend Ha Long Hotel, Ha Long Bay, Vietnam

Tutorials: Monday, 14 September 2026 (free for all AICAS participants) Venue: 144 Xuan-Thuy Road, Cau Giay Ward, Hanoi, Vietnam		
9:00 AM – 9:15 AM	Opening Session Room: 1st Floor, Jonathan KS Choi Culture Center, VNU Hanoi	
9:15 AM – 6:00 PM	Tutorial #1: AI-Enabled IC Design Automation and Intelligent Hardware Architectures Chairs: Minh-Thuy Le, HUST, Vietnam Quang-Kien Trinh, LQDTU, Vietnam Room: 1 st Floor, Jonathan KS Choi Culture Center, VNU Hanoi	Tutorial #2: High-Speed Interconnect and Clock Circuits for AI Computing Infrastructures Chairs: Du Yuan, Nanjing University, China; Duc-Tho Mai, ACT, Vietnam Room: 2nd Floor, Jonathan KS Choi Culture Center, VNU Hanoi
9:15 AM – 10:15 AM	Hardware-Assisted Verification for Next-Generation AI Hardware Accelerators Ibrahim Elfadel, <i>Khalifa University, United Arab Emirates</i>	Introduction to High-Speed SerDes Communications Quan Pan, <i>Southern University of Science and Technology, China</i>
10:15 AM	Tea Break (15 minutes)	
10:30 AM – 11:30 AM	AI-Driven Shift-Left for Circuit and System Design Xinfei Guo, <i>Shanghai Jiao Tong University, China</i>	High-Speed Ultra-Low Latency Mixed-Signal SerDes Transceiver Design Techniques for AI Computing Infrastructures Xiaoyan Gui, <i>Xi'an Jiaotong University, China</i>
11:45 AM	Lunch	
1:15 PM – 2:15 PM	AI for Analog IC Design Automation: Evolution and Open Challenges Nuno Horta, <i>University of Lisbon, Portugal</i>	Clock Generation Techniques for High-Speed Wireline Transceivers Hao Xu, <i>Fudan University, China</i>
2:15 PM – 3:15 PM	Bio-inspired Neuromorphic Olfaction: From Algorithms to Chips Hong Chen, <i>Tsinghua University, China</i>	Clock and Data Synchronization Techniques for High-Performance Interconnection Xiaoteng Zhao, <i>Xidian University, China</i>
3:15 PM	Tea Break (15 minutes)	
3:30 PM – 4:30 PM	Product-Ready Neuromorphic MCU at the Sensor Edge Amir Zjajo, <i>Innatera Nanosystems B.V., the Netherlands</i>	Design of Power-Efficient Reference-less PAM4 CDR for Ultra-High-Speed Wireline Communication Zhao Zhang, <i>Institute of Semiconductors, Chinese Academy of Sciences, China</i>

4:30 PM – 5:30 PM	Reconfigurable Accelerator Design for Multi-Precision Floating-Point Computing: From Processing Elements to SME-Oriented Architectures <i>Wei Mao, Xidian University, China</i>	Optical Receiver Front-Ends for AI Interconnect: Scenarios, Tradeoffs, and Design Guidelines <i>Dan Li, Xi'an Jiaotong University, China</i>
5:30 PM – 6:00 PM	AI-Driven EDA: Reshaping Chip Design in the AI Era <i>Tran Thuy Vy, Cadence Vietnam</i>	
Tuesday, 15 September 2026 Moving to Ha Long Bay Option 1: Free shuttle bus departure: 2:00 PM at 144 Xuan Thuy Road, Cau Giay Ward, Hanoi (advanced registration is required) Option 2: Day trip to visit Ha Long Bay on Cruise (participants pay the fee to our travel partner); hotel pickup at 7:15-8:00 AM		

Day 1: Wednesday, 16 September 2026 Venue: Wyndham Legend Halong Hotel Address: 12 Ha Long Street, Bai Chay Ward, Quang Ninh Province 02033				
9:00 AM – 9:30 AM	Conference Opening & GC/TPC Reporting			
9:30 AM – 10:30 AM	Keynote #1: From Design Automation to Autonomous Design: AI Super Agents for Chip Development Ziyad Hanna <i>Corporate VP, System Design Verification and AI Technologies at Cadence Design Systems</i> Venue: Diamond 1 Chairs: Du Li, Nanjing University, China			
10:30 AM	Tea Break (15 minutes)			
11:00 AM – 12:00 AM	Keynote #2: Learning from Biology - Neuromorphic Techniques for an Intelligent Vision System: Using Obstacle Avoidance as an Example Kea-Tiong (Samuel) Tang <i>National Tsing Hua University, Taiwan</i> Venue: Diamond 1 Chairs: Xuan-Tu Tran, Vietnam National University, Hanoi			
12:00 PM	Lunch			
1:30 PM – 2:45 PM	Session #1: AI Circuits and Systems for Robotics and Automation (4 papers) Venue: Panorama 1 Chairs: Yuhang Zhang, East China Normal University, China	Session #2: AI for Circuits and Systems I (5 papers) Venue: Panorama 2 Chairs: Jinming Lu, Nanjing University, China	Session #3: Emerging applications: IoT, healthcare, smart factories, environment, robotics, autonomous vehicles, drones, AR/VR, metaverse I (4 papers) Venue: Diamond 2 Chairs: Zhen Zhuang, The Chinese University of Hong Kong, China	Session #4: Circuits and systems for AI I (5 papers) Venue: Diamond 3 Chairs: HaiLong Zhao, PKU, China

2:45 PM	Tea Break (30 minutes)			
3:15 PM – 4:15 PM	Session #5: Computer vision algorithms and architectures I (4 papers) Venue: Panorama 1 Chairs: Duc-Tho Mai, Academy of Cryptography, Vietnam	Session #6 Special Session: AI-Powered Electronic Design Automation for IC Design I (4 papers) Venue: Panorama 2 Chairs: Bo Li, Xidian University, China	Session #7: AI for Future Wireless Communications (2 papers) Venue: Diamond 2 Chairs: Yuhang Zhang, East China Normal University, China	Session #8 Special Session: Emerging Energy-Efficient Hardware and Computing Paradigms for Edge Intelligence I (4 papers) Venue: Diamond 3 Chairs: Quang-Kien Trinh, LQDTU, Vietnam
4:15 PM	Tea Break (15 minutes)			
4:30 PM – 6:00 PM	Session #9: Architecture for AI computing I (5 papers) Venue: Panorama 1 Chairs: Zhou Wang, Imperial College London, Youngmin Kim, Hongik University, Korea	Session #10: Hardware/Software Co-Design and Design Automation for AI systems I (5 papers) Venue: Panorama 2 Chairs: Yichuan Bai, Nanjing University, China	Session #11: Emerging computing devices and materials for AI (6 papers) Venue: Diamond 2 Chairs: Hua Fan, UESTC, China	Session #12: In-Memory Computing I (5 papers) Venue: Diamond 3 Chairs: Bo Li, Xidian University, China
7:00 PM	Welcome reception			

Day 2: Thursday, 17 September 2026 Venue: Wyndham Legend Halong Hotel Address: 12 Ha Long Street, Bai Chay Ward, Quang Ninh Province 02033				
9:00 AM – 10:00 AM	Keynote #3: From FLOPS to Intelligence per Joule: The Case for Adaptive Dataflow Tulika Mitra <i>National University of Singapore, Singapore</i> Venue: Diamond 1 Chairs: Duy-Hieu Bui, VNU, Vietnam			
10:00 AM	Tea Break (30 minutes)			
10:30 AM – 11:30 AM	Keynote #4: From Intelligence to Infrastructure: Building the Global Ecosystem for the Next AI Era Hong-Viet Le <i>FPT Smart Cloud, Vietnam</i> Venue: Diamond 1 Chairs: Duy-Hieu Bui, Vietnam National University, Hanoi			
11:45 AM	Lunch			
1:15 PM – 2:30 PM	Session #13: Architecture for AI computing II (5 papers) Venue: Panorama 1 Chairs: Thu-Huong Truong, HUST, Vietnam	Session #14: AI for Circuits and Systems II (5 papers) Venue: Panorama 2 Chairs: Jinming Lu, Nanjing University, China	Session #15: Circuits and Systems for AI II (5 papers) Venue: Diamond 2 Chairs: Manh-Hiep Dao, VNU, Hanoi	Session #16 Special Session: Specialized Hardware and Systems for Edge Large Language Model Acceleration and Application, Precision-Scalable and Reliability- Aware AI Circuits for Closed-Loop Edge Autonomy, Multimedia Systems and Applications for AI (4 papers) Venue: Diamond 3 Chairs: Duy-Hieu Bui, VNU, Hanoi
2:30 PM	Tea Break (30 minutes)			

3:00 PM – 4:00 PM	Session #17: Computer vision algorithms and architectures II (3 papers) Venue: Panorama 1 Chairs: Qing Zhang, Shenzhen University, China	Session #18 Special Session: AI-Powered Electronic Design Automation for IC Design II (4 papers) Venue: Panorama 2 Chairs: Thanh-Loan Pham Nguyen, HUST, Vietnam	Session #19 Special Session: Emerging Energy-Efficient Hardware and Computing Paradigms for Edge Intelligence II (4 papers) Venue: Diamond 2 Chairs: Xueqing Li, Tsinghua University, China	Grand Challenge SS'25 Venue: Diamond 3 Chairs: Wei Mao, Xidian University, China
4:00 PM	Tea Break (15 minutes)			
4:15 PM – 5:45 PM	Session #20: Emerging applications: IoT, healthcare, smart factories, environment, robotics, autonomous vehicles, drones, AR/VR, metaverse II (4 papers) Venue: Panorama 1 Chairs: Manh-Hiep Dao, VNU Hanoi	Session #21: Hardware/Software Co-Design and Design Automation for AI systems II (5 papers) Venue: Panorama 2 Chairs: Zhen Zhuang, The Chinese University of Hong Kong	Session #22 Special Session: Compute-in-Memory Co-Design: From Emerging Devices to AI Application Optimization (5 papers) Venue: Diamond 2 Chairs: Quoc-Hung Nguyen, VNU, Hanoi	Grand Challenge SS'26 Venue: Diamond 3 Chairs: Wei Mao, Xidian University, China
7:00 PM	Banquet & Award Ceremony			

Day 3: Friday, 18 September 2026 Venue: Wyndham Legend Halong Hotel Address: 12 Ha Long Street, Bai Chay Ward, Quang Ninh Province 02033		
9:00 AM – 10:30 AM	Session #23: Neuromorphic and bioinspired circuits, processors, systems, and their applications I (5 papers) Venue: Panorama 1 Chairs: Manh-Hiep Dao, VNU, Hanoi	Session #24: In-Memory Computing II (6 papers) Venue: Panorama 2 Chairs: Anh Tuan Do, A*STAR, Singapore
10:30 AM	Tea Break (15 minutes)	
10:45 AM – 11:45 AM	Session #25: Neuromorphic and bioinspired circuits, processors, systems, and their applications II (6 papers) Venue: Panorama 1 Chairs: Duc-Tho Mai, ACT, Vietnam	Session #26 Special Session: Compute-in-Memory Co-Design: From Emerging Devices to AI Application Optimization II (4 papers) Venue: Panorama 2 Chairs: Qing Zhang, Shenzhen University, China
12:15 PM	Lunch	
1:30 PM – 5:15 PM	Workshop #1: The 2nd IEEE Chiplet Workshop on Circuits and Systems for Next-Generation AI Computing Venue: Panorama 1 Chairs: Chixiao Chen, Fudan University, China; Anh Tuan Do, A*STAR, Singapore.	Industry talk: HiFloat Data Format Principles, Practices and Standards Haonan Zhang, <i>Huawei, China</i> Venue: Panorama 2 Chairs: Manh-Hiep Dao, VNU, Hanoi
		Workshop #2: Smart Sensor Circuit and Intelligent Systems for Emerging Applications Venue: Panorama 2 Chairs: Qing Zhang, Shenzhen University, China; Milin Zhang, Tsinghua University, China
5:15 PM	Closing	

Technical Program

Electronic version: <https://2026.ieee-aicas.org/program/>



Wednesday, September 16, 2026

Day 1: Wednesday, 16 September 2026 Venue: Wyndham Legend Halong Hotel Address: 12 Ha Long Street, Bai Chay Ward, Quang Ninh Province 02033				
8:30 AM – 5:00 PM	Live demonstration & Exhibits			
Time	Paper #	Title	Authors	Type
Whole day	1571274451	Live Demonstration: A Hardware-Efficient Arc Early Detection System via DWT-ConvLSTM at Edge	Zhenhao Wang, Xuhui Du and Jialong Wang (Xidian University, China); Fan Jiang and Xian Ding (Beijing Smart-Chip Microelectronics Technology Co., Ltd); Xiaoxi Li, Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Exhibit
Whole day	1571274667	Live Demonstration: Edge-Deployed Compound AI with RAG for High-Accuracy Embodied Robotic Planning	Yitong Shen, Mingde Duan, Zhenhao Wang, Yingjie Zhao, Yanwen Ding and Xuhui Du (Xidian University, China); Sai Zhang (Junhe Technology (Quzhou) Co., Ltd.); Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Exhibit
Whole day	1571306695	Live Demonstration: A Real-Time Reservoir Processor on FPGA for Edge Temporal Prediction	Yucen Shi, Yuncheng Lu, Zehao Li, Junying Li and Tony Tae Hyoung Kim (Nanyang Technological University, Singapore)	Exhibit
Whole day	-	Huawei exhibition	Haonan Zhang (Huawei, China)	Exhibit
Whole day	-	CASS Vietnam Chapter	Manh-Hiep Dao, Vietnam National University, Hanoi	Exhibit

9:00 AM — 9:30 AM	Conference Opening & GC/TPC Reporting
9:30 AM — 10:30 AM	Keynote #1: From Design Automation to Autonomous Design: AI Super Agents for Chip Development Ziyad Hanna Corporate VP, System Design Verification and AI Technologies at Cadence Design Systems Venue: Diamond 1 Chairs: Du Li, Nanjing University, China
10:30 AM	Tea break
11:00 AM — 12:00 PM	Keynote #2: Learning from Biology - Neuromorphic Techniques for an Intelligent Vision System: Using Obstacle Avoidance as an Example Kea-Tiong (Samuel) Tang National Tsing Hua University, Taiwan Venue: Diamond 1 Chairs: Xuan-Tu Tran, Vietnam National University, Hanoi
12:00 PM	Lunch

1:30 PM– 2:45 PM	Session #1: AI Circuits and Systems for Robotics and Automation (4 papers) Venue: Panorama 1 Chairs: Yuhang Zhang, East China Normal University, China			
Time	Paper #	Title	Authors	Type
1:30 PM	1571268204	A 40 μ A, 1-14 V Load-Adaptive in-Situ Neural Stimulator	Dawei Li, Yao Zhou and Ziyu Chen (South-Central Minzu University, China); Dongqin Deng (Wuhan University, China); Li Zhang (Hubei University of Technology, China); Yuan Du and Li Du (Nanjing University, China); Dongsheng Liu (Huazhong University of Science and Technology, China)	Oral
1:45 PM	1571267971	GAP-KV: Granularity-Aware Pruning for Chunk-Level KV Cache Compression in Long-Context LLMs	Hakyeong Lee, Sangbeom Jeong and Hyun Kim (Seoul National University of Science and Technology, Korea (South))	Oral
2:00 PM	1571268277	Lightweight Pallet Detection and Pose Estimation Technology for Pallet Truck Automated Guided Vehicles	Jize Liu, Zherong Liu (University of Electronic Science and Technology of China (UESTC), China); Tianyi Zhao (Chengdu Ruixinxing Technology Co., Ltd., China); Wentao Fan (University of Electronic Science and Technology of China (UESTC), China); Jiayao Huang (Chengdu Ruixinxing Technology Co., Ltd., China); Hang Chen, Haoxiang Long and Bozhang Hua (University of Electronic Science and Technology of China (UESTC), China); Yu Long and Ling Xu (Chengdu Ruixinxing Technology Co., Ltd., China); Ye Liu and Jun Zhou (University of Electronic Science and Technology of China (UESTC), China);	Oral
2:15 PM	1571274040	Embedded AI Architecture Evaluation for in-Barn Real-Time Bovine Mastitis Detection at the Edge	François Gonothi Toure, Abdoulaye Baniré Diallo and Mounir Boukadoum (Université du Québec à Montréal, Canada)	Oral

1:30 PM– 2:45 PM	Session #2: AI for Circuits and Systems I (5 papers) Venue: Panorama 2 Chairs: Jinming Lu, Nanjing University, China			
Time	Paper #	Title	Authors	Type
1:30 PM	1571252614	Beyond Iterative Search: Intelligent Generative GATv2 Framework for Analog Sizing	Fan Hu, Shaik Jani Babu and Xinfei Guo (Shanghai Jiao Tong University, China)	Oral
1:45 PM	1571264263	Curriculum Learning as an Enabling Methodology for Reinforcement Learning in Analog Circuit Sizing	Douglas Mateus Machado, Nathan Chanez and Severin Trochut (STMicroelectronics, France); Paul Chollet (Télécom Paris, France); Roberto Guizzetti and Pascal Urard (STMicroelectronics, France); Van-Tam Nguyen (Télécom Paris, France)	Oral
2:00 PM	1571268026	20 Hours from Spec to Schematic: A Fully Automated Design Framework for Continuous-Time Sigma-Delta Modulators	Yuxuan Zhang, Chen Wang, Zhuohua Liu and Yuanqi Hu (Beihang University, China); Wei W. Xing (University of Sheffield, United Kingdom (Great Britain))	Oral
2:15 PM	1571268245	PIECE: Projection-Invariant Normalization for Energy-Aware Compression in Efficient Transformer Inference on Chiplets	Tzu-Ching Yang, Jiing-Ping Wang and An-Yeu (Andy) Wu (National Taiwan University, Taiwan)	Oral
2:30 PM	1571268458	Multimodal Graph-Image Reinforcement Learning for VLSI Macro Placement	Yi-Syuan Shih (National Central University, Taiwan); Yu-Guang Chen (National Tsing Hua University, Taiwan); Jia-Sin Pan (National Central University, Taiwan); Shih-Cheng Huang, Cheng-Hong Tsai and De-Shiun Fu (Global Unichip Corporation, Taiwan); Mango Chia-Tso Chao (National Yang Ming Chiao Tung University, Taiwan)	Oral

1:30 PM– 2:45 PM	Session #3: Emerging applications: IoT, healthcare, smart factories, environment, robotics, autonomous vehicles, drones, AR/VR, metaverse I (4 papers) Venue: Diamond 2 Chairs: Zhen Zhuang, The Chinese University of Hong Kong, China			
Time	Paper #	Title	Authors	Type
1:30 PM	1571268406	PanLUNA: An Efficient and Robust Query-Unified Multimodal Model for Edge Biosignal Intelligence	Marija Zelic; Anna Tegon; Yawei Li; Thorir Mar Ingolfsson; Luca Benini (ETH Zürich, Switzerland)	Oral
1:45 PM	1571250142	MAPLE: Efficient Agglayer Proving via off-Chain Pre-Verification for Metaverse Multi-Chain Systems	Huyen Nguyen, Lan Ngoc Le, Khoa Tan VO, Thu Nguyen and Tu-Anh Nguyen-Hoang (University of Information Technology - VNU-HCM, Vietnam)	Oral
2:00 PM	1571264335	Adaptive Routing for Efficient Diffusion Transformer-Based PNI Prediction	Youngun Han (Seoul National University, Korea (South) & OUTTA, Korea (South)); Dohyun Kweon (Kyung Hee University, Korea (South)); Kyeonghun Kim (OUTTA, Korea (South)); Hyunsu Go, Jina Jeong and Suah Park (Seoul National University, Korea (South)); Induk Um (Chung-Ang University, Korea (South)); Junga Kim, Anna Jung and Yulji Jeong (Seoul National University, Korea (South)); Sungja Park (Seoul National University School of Medicine, Korea (South) & Seoul National University, Korea (South)); Jinyong Jun (Seoul National University, Korea (South)); Pa Hong (Samsung Changwon Hospital, Korea (South)); Woo Kyoung Jeong (Samsung Medical Center, Korea (South)); Won Jae Lee (Samsung Medical Center, Sungkyunkwan University School of Medicine, Korea (South)); Ken Ying-Kai Liao (NVIDIA, Taiwan); Hyuk-Jae Lee and Nam-Joon Kim (Seoul National University, Korea (South))	Oral
2:15 PM	1571264882	HLCFN-PAD: Hybrid Loss-CBAM Model for Partial Fingerprint Anti-Spoofing	Dao-Heng Luo and Ching-Te Chiu (National Tsing Hua University, Taiwan); Ned Lin, Ang Yu, Paul Lai and Daniel Chen (FocalTech Electronics Co., Ltd., Taiwan); Shao-En Lin (National Tsinghua University, Taiwan)	Oral

1:30 PM– 2:45 PM	Session #4: Circuits and systems for AI I (5 papers) Venue: Diamond 3 Chairs: HaiLong Zhao, PKU, China			
Time	Paper #	Title	Authors	Type
1:30 PM	1571263159	PHSim: An Efficient Simulator for LLM Inference on PIM-Based Heterogeneous Systems	Yueqi Zhang (Nanjing University, China); Chao Fang (KU Leuven, Belgium); Xianshi Jiang, Jie Chen and Yichuan Bai (Nanjing University, China); Peng Qin, Dawei Wang and Xu Zhang (Intel Labs, China); Yuan Du and Li Du (Nanjing University, China)	Oral
1:45 PM	1571265653	AP-Rec: Adaptive Dual-Path Inference for Scalable Multi-GPU Recommendation Systems	Jiyoun Hwang, Jongmin Seok and Chae Eun Rhee (Hanyang University, Korea (South))	Oral
2:00 PM	1571267418	An Accurate Yet Compact Softmax Hardware Implementation Optimized for Microscaling Data Format	Meng-Hsun Hsieh and Juinn-Dar Huang (National Yang Ming Chiao Tung University, Taiwan)	Oral
2:15 PM	1571268062	A Lightweight Accelerator for Kolmogorov-Arnold Transformers in Edge Inference	Chengxing Fan, Xuliang Yu and Jie Hu (Zhejiang University, China); Wei-Han Yu (University of Macau, Macao); Liang Zhao (Zhejiang University, China)	Oral
2:30 PM	1571268419	AMBER: Thermal-Aware Spatio-Temporal Scheduling for Edge LLM Inference on 3D DRAM-on-NPU Systems	Xuanda Lin, Siyao Jia and Haozhe Zhu (Fudan University, China); Xinru Jia (CiMicro.AI, China); Shiwei Liu (Tongji University); Chixiao Chen (Fudan University, China)	Oral
2:45 PM	Tea break			

3:15 PM– 4:15 PM	Session #5: Computer vision algorithms and architectures I (4 papers) Venue: Panorama 1 Chairs: Duc-Tho Mai, Academy of Cryptography, Vietnam			
Time	Paper #	Title	Authors	Type
3:15 PM	1571254629	ADORE-VLM: Attention-Dynamics-Oriented Re-Entry-Based Early Exiting for Accelerating Vision-Language Model Inference	Tackwoo Nam, Gilha Lee and Hyun Kim (Seoul National University of Science and Technology, Korea (South))	Oral
3:30 PM	1571264329	Anatomy-Privileged Distillation with Token Routing for MRI-Based Prediction of Perineural Invasion	Hyunsu Go (Seoul National University, Korea (South)); Youngung Han (Seoul National University, Korea (South)) & OUTTA, Korea (South)); Kyeonghun Kim (OUTTA, Korea (South)); Junga Kim (Seoul National University, Korea (South)); Dohyun Kweon (Kyung Hee University, Korea (South)); Jinyong Jun (Seoul National University, Korea (South)); Sungha Park (Seoul National University School of Medicine, Korea (South)) & Seoul National University, Korea (South)); Anna Jung (Seoul National University, Korea (South)); Induk Um (Chung-Ang University, Korea (South)); Yulji Jeong, Suah Park, Jina Jeong and Suemin Yang (Seoul National University, Korea (South)); Pa Hong (Samsung Changwon Hospital, Korea (South)); Woo Kyoung Jeong (Samsung Medical Center, Korea (South)); Won Jae Lee (Samsung Medical Center, Sungkyunkwan University School of Medicine, Korea (South)); Ken Ying-Kai Liao (NVIDIA, Taiwan); Hyuk-Jae Lee and Nam-Joon Kim (Seoul National University, Korea (South))	Oral
3:45 PM	1571264333	SpikeDS: Dual Sparsity Spikformer for Perineural Invasion Prediction in 3D MRI	Induk Um (Chung-Ang University, Korea (South)); Youngung Han (Seoul National University, Korea (South)) & OUTTA, Korea (South)); Kyeonghun Kim (OUTTA, Korea (South)); Yulji Jeong, Jina Jeong and Hyunsu Go (Seoul National University, Korea (South)); Dohyun Kweon (Kyung Hee University, Korea (South)); Sungha Park (Seoul National University School of Medicine, Korea (South)) & Seoul National University, Korea (South)); Junga Kim, Anna Jung, Suah Park and Hyuk-Jae Lee (Seoul National University, Korea (South)); Pa Hong (Samsung Changwon Hospital, Korea (South)); Woo Kyoung Jeong (Samsung Medical Center, Korea (South)); Won Jae Lee (Samsung Medical Center, Sungkyunkwan University School of Medicine, Korea (South)); Ken Ying-Kai Liao (NVIDIA, Taiwan); Nam-Joon Kim (Seoul National University, Korea (South))	Oral
4:00 PM	1571267460	Robust Image Classification Under Complex Illumination: A FastViT Framework with Sharpness-Aware Minimization	Bohan Guo, Chaoyao Shen, Yuning Ji, Yu Liang and Meng Zhang (Southeast University, China)	Oral

3:15 PM– 4:15 PM	Session #6 Special Session: AI-Powered Electronic Design Automation for IC Design I (4 papers) Venue: Panorama 2 Chairs: Bo Li, Xidian University, China			
Time	Paper #	Title	Authors	Type
3:15 PM	1571266792	SafeRTL: IP-Preserving Verilog Optimization via Principle-Guided Local-Cloud LLM Framework	Jing Wang (Nanjing University of Posts and Telecommunications, China); Zheng Li (Fudan University, China); Lei Li (The University of Hong Kong, Hong Kong); Fan He and Liyu Lin (Fudan University, China); Yao Lai (The University of Hong Kong, Hong Kong); Jiafei Yao (Nanjing University of Posts and Telecommunications, China); Yan Li and Xiaoyang Zeng (Fudan University, China); Yufeng Guo (Nanjing University of Posts and Telecommunications, China)	Oral
3:30 PM	1571268118	Efficient Machine Learning-Driven RF Post-Layout Prediction via Bidirectional State-Space Vision Mamba	Hoang Trong Nguyen, Lihong Zhang and Trung Q. Duong (Memorial University of Newfoundland, Canada)	Oral
3:45 PM	1571268248	AI Chip Design Tool with Self-Learning Capability for FinFET Advanced Processes	Dan Liu, Qian Wu and Xiaoshi Zhao (Beijing Tango Software Technology Company, China)	Oral
4:00 PM	1571268363	Chiplet-NSRL: A Neuro-Symbolic Reinforcement Learning Approach for Interpretable Design Space Exploration of Chiplet-Based AI Accelerator	Ruifeng Zhang, Bo Li, Xiguang Wu and Wei Mao (Xidian University, China); Guoyong Shi (Shanghai Jiao Tong University, China); Yan Liu and Genquan Han (Xidian University, China)	Oral

3:15 PM– 4:00 PM	Session #7: AI for Future Wireless Communications (2 papers) Venue: Diamond 2 Chairs: Yuhang Zhang, East China Normal University, China			
Time	Paper #	Title	Authors	Type
3:15 PM	1571268896	Digital Predistortion Design with Lightweight Variation-Aware Network for Sub-THz Band	Tian Sheuan Chang and Tzu-Chia Tseng (National Yang Ming Chiao Tung University, Taiwan)	Oral
3:30 PM	1571272151	A Binary Matrix Optimization Framework for the Automated Design of Broadband Microstrip Resonators	Fu Qi Chua (Universiti Malaya, Malaysia); Chao Wang and Jinghua Zhang (Shanghai Jiao Tong University, China); Kian Kit Cheah and Harikrishnan Ramiah (Universiti Malaya, Malaysia); Yongfu Li (Shanghai Jiao Tong University, China)	Oral
3:15 PM– 4:15 PM	Session #8 Special Session: Emerging Energy-Efficient Hardware and Computing Paradigms for Edge Intelligence I (4 papers) Venue: Diamond 3 Chairs: Quang-Kien Trinh, LQDTU, Vietnam			
Time	Paper #	Title	Authors	Type
3:15 PM	1571268395	A Novel Delta-Sigma Computing-in-Memory Macro Based on FeRAM for Edge Computing	Huanning Zhang, Bo Li, Xiguang Wu, Wei Mao, Xiao Yu, Bing Chen, Yan Liu and Genquan Han (Xidian University, China)	Oral
3:30 PM	1571268472	SpikeLogBERT: Energy-Efficient Log Parsing Using Spiking Transformer Networks	Thuan Bui and Duong Do (Swinburne Vietnam, FPT University, Vietnam); Tung Vu (Posts and Telecommunications Institute of Technology, Vietnam); Duc-Tho Mai (Academy of Cryptography Techniques, Vietnam); Cong-Kha Pham (University of Electro-Communications (UEC), Japan)	Oral
3:45 PM	1571268951	A Real-Time Vietnamese Speech Acquisition and Recognition System for Control Applications on FPGA Technology	Dang-Quang Nguyen, The-Anh Nguyen, Duy-Hieu Bui and Xuan-Tu Tran (Information Technology Institute, Vietnam National University, Hanoi)	Oral
4:00 PM	1571273751	MRTPE: A 28nm 131.76 TOPS/W and 8 Core RISC-V DNN Inference Processor with Multiple RISC-V Transformational Mechanisms	Zhou Wang (Imperial Global Singapore, Imperial College London); Jiuren Zhou and Xiguang Wu (Xidian University, China); Xiaonan Tang (Wisemaytech, China); Shushan Qiao (Institute of Microelectronics of the Chinese Academy of Sciences, China); Tianchun Ye (IME-CAS, China); Anil A Bharath (Imperial College & Cortextica Vision Systems, United Kingdom (Great Britain)); Emm Mic Drakakis (Imperial College London, United Kingdom (Great Britain))	Oral

4:15 PM	Tea break			
4:30 PM– 5:45 PM	Session #9: Architecture for AI computing I (5 papers) Venue: Panorama 1 Chairs: Zhou Wang, Imperial Global Singapore of Imperial College London, Youngmin Kim, Hongik University, Korea			
Time	Paper #	Title	Authors	Type
4:30 PM	1571264636	KV Cache Management for Block Diffusion Language Models on Multi-Level NV-Memory	Masafumi Higashi, Shota Suzuki, Yuya Yamamoto, Naoko Misawa, Chihiro Matsui and Ken Takeuchi (The University of Tokyo, Japan)	Oral
4:45 PM	1571268263	A Low-Power Sparse Convolution Accelerator with Idle-First-Task-Assignment for Edge Vision	Jingyue Zhuge (TUD Dresden University of Technology, Germany & ScaDS.AI, Germany); Johannes Partzsch (TUD Dresden University of Technology, Germany); Christian Mayr (TU Dresden, Germany)	Oral
5:00 PM	1571268280	Sub-100KB on-Device CNN Training via Dynamic Sparse Updates and Hardware--Software Co-Design	Tian Sheuan Chang and I-Hsuan Li (National Yang Ming Chiao Tung University, Taiwan)	Oral
5:15 PM	1571268292	A Multi-Precision Accelerator for Partial Differential Equation Solving with Collaborative Physics-Informed Neural Networks and Finite Difference Methods	Yichuan Bai, Botao Wu and Xiaoyu Chang (Nanjing University, China); Chongkang Tan (Ant Group, China); Yuan Du and Li Du (Nanjing University, China)	Oral
5:30 PM	1571274252	When NPUs are Not Always Faster: A Stage-Level Analysis of Mobile LLM Inference	Pu Li, Jiawen Qi and Qinyu Chen (Leiden University, The Netherlands)	Oral

4:30 PM– 5:45 PM	Session #10: Hardware/Software Co-Design and Design Automation for AI systems I (5 papers) Venue: Panorama 2 Chairs: Yichuan Bai, Nanjing University, China			
Time	Paper #	Title	Authors	Type
4:30 PM	1571265285	DISA: Algorithm-Hardware Co-Design of a Sparsity-Induced Deep Unfolding Network for Edge Compressive Sensing	Xuewei Cao, Wenbiao Zhou, Fengchi Lu, Wenxuan Niu and Zhihan Zhang (Beijing Institute of Technology, China)	Oral
4:45 PM	1571266794	SHIELD: A Segmented Hierarchical Memory Architecture for Energy-Efficient LLM Inference on Edge NPUs	Jintao Zhang and Xuanyao Fong (National University of Singapore, Singapore)	Oral
5:00 PM	1571267994	NPN-ApproxLib: DNN-Aware Approximate Multiplier Design via NPN Classes	Sallar Ahmadi-Pour, Chandan Kumar Jha and Sajjad Parvin (University of Bremen, Germany); Rolf Drechsler (University of Bremen/DFKI, Germany)	Oral
5:15 PM	1571268453	DiffABCO: Shift-Left Wafer-Scale Integration Design via Differentiable Architecture-Backend Co-Optimization	Zhen Zhuang (Chinese University of Hong Kong, Hong Kong); Zheng Yang (Georgia Institute of Technology, USA); Kai-Yuan Chao (Siemens Digital Industries Software, USA); Tsung-Yi Ho (The Chinese University of Hong Kong, Hong Kong)	Oral
5:30 PM	1571268123	SNN2Bitstream: Automated Model-to-FPGA HLS-Based Framework for Spiking Neural Network Hardware Accelerators	Xindan Zhang, Ioannis Kaskampas, Andrea Pinna and Haralampos-G. Stratigopoulos (Sorbonne Université, CNRS, LIP6, France)	Oral

4:30 PM– 5:45 PM	Session #11: Emerging computing devices and materials for AI (6 papers) Venue: Diamond 2 Chairs: Hua Fan, UESTC, China			
Time	Paper #	Title	Authors	Type
4:30 PM	1571268285	Dual-Channel-Shared OFA-p-SOT-MRAM CIM Macro for High-Throughput Binary Similarity Search in LLM KV-Cache Compression	Fantao Gao, Cancheng Xiao, Dingsong Jiang, Xun Tang and Jiahao Zhao (Tsinghua University, China); Tao Yang (ENFI Engineering Corporation, China); Jianshi Tang (Tsinghua University, China); Tianxiang Nan (School of Integrated Circuits, Tsinghua University, China)	Oral
4:45 PM	1571274339	Neural Sequence Models for Memristor State Variable Modeling and Prediction	Rishabh Mallik and Ming-Jay Yang, Stephan Menzel, Emre Neftci, John Paul Strachan (Peter Grünberg Institute, Forschungszentrum Jülich, Germany)	Oral
5:00 PM	1571264447	TinyGLASS: Real-Time Self-Supervised In-Sensor Anomaly Detection	Pietro Bonazzi (ETH Zurich, Switzerland); Rafael Sutter (ETH Zürich, Switzerland); Luigi Capogrosso (Interdisciplinary Transformation University of Austria, Austria); Mischa Buob (Swiss Engineering Partners AG, Switzerland); Michele Magno (ETH Zurich, Switzerland)	Oral
5:15 PM	1571268343	A Transposable VC-SOT-MRAM CIM Macro for Edge AI Tasks with High Linearity	Bingqian Song, Cancheng Xiao, Yiyuan Xiao, En Li, Jiahao Zhao, Yining Li, Yuheng Yang and Jianshi Tang (Tsinghua University, China); Qi Hu (Beijing Superstring Academy of Memory Technology, China); Tianxiang Nan (School of Integrated Circuits, Tsinghua University, China)	Oral
5:30 PM	1571268494	A Hardware-Oriented Event-Driven Gesture Recognition System with a Geometry-Token Early-Exit Classifier for a Perovskite Vision Platform	Junying Li and Chufeng Yang (Nanyang Technological University, Singapore); Kyung-Chan An (Seokyeong University, Korea (South)); Yuncheng Lu and Kiho Seong (Nanyang Technological University, Singapore); Ng Si En Timothy (Stanford University, USA); Shibi Varku, Natalia Yantara and Qinjie Wu (Nanyang Technological University, Singapore); Arindam Basu (City University of Hong Kong, Germany); Nripan Mathews and Tony Tae Hyoung Kim (Nanyang Technological University, Singapore)	Oral
5:45 PM	1571268512	3D Monolithic Integrated SNN for Ultra-Low Energy ECG Arrhythmia Classification	Shrees Shrinivas Nagral (Indian Institute of Technology, Bombay, India); Harshitha Gangu, Laxmeesha Somappa and Veeresh Deshpande (Indian Institute of Technology Bombay, India)	Oral

4:30 PM– 5:45 PM	Session #12: In-Memory Computing I (5 papers) Venue: Diamond 3 Chairs: Bo Li, Xidian University, China			
Time	Paper #	Title	Authors	Type
4:30 PM	1571265000	Analysis of Accuracy, Precision, and Error Sources in Bitline Voltage Regulator for AIMC Accelerators	Berk Bozkurt and Riccardo Vignali (University of Pavia, Italy); Riccardo Zurla (STMicroelectronics, Pavia, Italy); Marco Pasotti (STMicroelectronics, Agrate Brianza, Italy); Andrea Lico and Alessio Antolini (ARCES-DEI, University of Bologna, Italy); Eleonora Franchi (University of Bologna, Italy); Alessandro Cabrini (University of Pavia, Italy)	Oral
4:45 PM	1571266241	A 15.7TOPS/W NAND Flash-Based Process-in-Memory Featuring a Time-Domain Computation Unit	Hyuk-Jin Lee and Kyunghoon Pyo (Korea Advanced Institute of Science & Technology, Korea (South)); SeongHwan Cho (Korea Advanced Institute of Science and Technology (KAIST), Korea (South))	Oral
5:00 PM	1571267829	NV-Memory Centric Liquid Neural Network Accelerator for Reflexive Robotic Control	Yuya Yamamoto, Naoko Misawa, Chihiro Matsui and Ken Takeuchi (The University of Tokyo, Japan)	Oral
5:15 PM	1571268180	PSD: Pairwise Signed-Digit for SRAM-Based Digital Compute-in-Memory with OR-Approximated Adder Tree	Jiing-Ping Wang and An-Yeu Wu (National Taiwan University, Taiwan)	Oral
5:30 PM	1571268234	An Energy-Efficient Multi-Edge Multi-Input Delay Cell with Asymmetric Parasitic Compensation for TD-CIM	Junyeong Heo, Joo Hyeon Jeon, Sung In Cho and Jinwook Burm (Sogang University, Korea (South))	Oral
7:00 PM	Welcome reception Shuttle buses will depart from the conference venue (the hotel's lobby) at 18:00 to the restaurant.			

Thursday, September 17, 2026

Day 2: Thursday, 17 September 2026 Venue: Wyndham Legend Halong Hotel Address: 12 Ha Long Street, Bai Chay Ward, Quang Ninh Province 02033				
8:30 AM–5:00 PM	Live demonstration & Exhibits			
Time	Paper #	Title	Authors	Type
Whole day	1571274451	Live Demonstration: A Hardware-Efficient Arc Early Detection System via DWT-ConvLSTM at Edge	Zhenhao Wang, Xuhui Du and Jialong Wang (Xidian University, China); Fan Jiang and Xian Ding (Beijing Smart-Chip Microelectronics Technology Co., Ltd); Xiaoxi Li, Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Exhibit
Whole day	1571274667	Live Demonstration: Edge-Deployed Compound AI with RAG for High-Accuracy Embodied Robotic Planning	Yitong Shen, Mingde Duan, Zhenhao Wang, Yingjie Zhao, Yanwen Ding and Xuhui Du (Xidian University, China); Sai Zhang (Junhe Technology (Quzhou) Co., Ltd.); Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Exhibit
Whole day	1571306695	Live Demonstration: A Real-Time Reservoir Processor on FPGA for Edge Temporal Prediction	Yucen Shi, Yuncheng Lu, Zehao Li, Junying Li and Tony Tae Hyoung Kim (Nanyang Technological University, Singapore)	Exhibit
Whole day	-	Huawei exhibition	Haonan Zhang (Huawei, China)	Exhibit
Whole day	-	CASS Vietnam Chapter	Manh-Hiep Dao, Vietnam National University, Hanoi	Exhibit

9:00 AM – 10:00 AM	Keynote #3: From FLOPS to Intelligence per Joule: The Case for Adaptive Dataflow Tulika Mitra National University of Singapore, Singapore Venue: Diamond 1 Chairs: Duy-Hieu Bui, VNU, Vietnam
10:00 AM	Tea break
10:30 AM – 11:30 AM	Keynote #4: From Intelligence to Infrastructure: Building the Global Ecosystem for the Next AI Era Hong-Viet Le FPT Smart Cloud, Vietnam Venue: Diamond 1 Chairs: Duy-Hieu Bui, Vietnam National University, Hanoi
11:45 AM	Lunch

1:15 PM– 2:30 PM	Session #13: Architecture for AI computing II (5 papers) Venue: Panorama 1 Chairs: Thu-Huong Truong, HUST, Vietnam			
Time	Paper #	Title	Authors	Type
1:15 PM	1571268455	Area- and Energy-Efficient Base-2 Softmax Design for DNN Inference with 1.95x Scaled Efficiency	Ting-Yi Chan, Hsiao-Jung Lin, Min-Yu Lin, Nidhi, and Jinn-Shyan Wang (National Chung Cheng University, Taiwan)	Oral
1:30 PM	1571268928	CRISP: Partitioning LLM Inference Across Main Memory and HBM via Input Sparsification and Low-Rank Approximation	Koki Shibata and Shota Suzuki (The University of Tokyo, Japan); Naoko Misawa (The University of Tokyo); Chihiro Matsui (The University of Tokyo, Japan); Ken Takeuchi (The University of Tokyo)	Oral
1:45 PM	1571273686	SARA: A Scalable ASIP-Based RISC-V Architecture for Edge AI Applications	Kuan-Wei Chen, Yu-Hao Cheng, Yu-Hsiang Lee and Hsie-Chia Chang (National Chiao Tung University, Taiwan)	Oral
2:15 PM	1571274664	Low-Power GELU-Induced Dynamic Precision for Feed-Forward Networks in Transformers	Kai-Chieh Hsu and Tian Sheuan Chang (National Yang Ming Chiao Tung University, Taiwan)	Oral

1:15 PM– 2:30 PM	Session #14: AI for Circuits and Systems II (5 papers) Venue: Panorama 2 Chairs: Jinming Lu, Nanjing University, China			
Time	Paper #	Title	Authors	Type
1:15 PM	1571268460	Distributed VLM and CNN Co-Processing for Edge AI over Non-Equalized RFSOC DAC-to-ADC Raw Channels	Muhammad Junaid, Dohyeon Kwon and Eva Cha (Daegu Gyeongbuk Institute of Science and Technology, Korea (South)); Korkut K Tokgöz (Sabanci University, Turkey & Evrim Co. Ltd., Japan); Gain Kim (DGIST, Korea (South))	Oral
1:30 PM	1571268502	Toward Resource-Efficient Spiking Neural Networks for MRI Classification	Khoa-Sang Nguyen and Quang-Kien Trinh (Le Quy Don Technical University, Vietnam); Deepu John (University College Dublin, Ireland); Dinh-Toan Le (108 Military Central Hospital, Vietnam); Thi-Thuy-Linh Le and Thi-Hong-Tham Tran (Le Quy Don Technical University, Vietnam)	Oral
1:45 PM	1571274172	Machine Learning-Assisted Design and Optimization of Tunable Bandpass LNA for Ultrasound Receiver Analog-Frontend	Xie Xinzhe (Nanyang Technological University, Singapore); Jinhai Hu (Institute of Microelectronics, Agency for Science, Technology and Research, Singapore); Zhaoyang Cao, Siqi Zhang, Jiaju Lu and Wang Ling Goh (Nanyang Technological University, Singapore); Litter Siek (Nanyang Technological University & Deputy Director, VIRTUS-IC Design Center of Excellence, Singapore); Abdulkadir C Yucel (Nanyang Technological University, Singapore); Yuan Gao (Institute of Microelectronics, A*STAR, Singapore)	Oral
2:00 PM	1571274661	InvCal-ITF: Fast and Accurate ITF Calibration via Surrogate-Assisted Inverse Optimization	Tianyi Xu and Hongwei Yao, Bingyi Ye and Yang Shen, Xiaojin Li, Yabin Sun and Yuhang Zhang (East China Normal University, China)	Oral
2:15 PM	1571274669	MonarchGRU: A Low-Footprint Recurrent Neural Network for Audio Keyword Spotting System	Soujanya Bhowmick (Aalto University, Finland); Sangyeob Kim (Yonsei University, Korea (South)); Qinyu Chen (Leiden University, The Netherlands); Chang Gao (Delft University of Technology, The Netherlands); Kwantae Kim (Aalto University, Finland)	Oral

1:15 PM– 2:30 PM	Session #15 Circuits and systems for AI II (5 papers) Venue: Diamond 2 Chairs: Manh-Hiep Dao, VNU, Hanoi			
Time	Paper #	Title	Authors	Type
1:15 PM	1571268251	CMOS Oscillator Cluster with Analog-Tunable Coupling for Frequency-Encoded Computing	Clément Dufossée, Jean-Baptiste Begueret, Thierry Taris, Sylvain Saïghi, François Sandrez and Adrien F. Vincent (University of Bordeaux, France)	Oral
1:30 PM	1571268488	HetKey: A Cross-Device Benchmark and Edge AI Pipeline for Acoustic Keystroke Attacks	Tuan-Duc Nguyen and Thuan Bui (Swinburne Vietnam, FPT University, Vietnam); Dinh-Dung Nguyen (Swinburne Vietnam, FPT University, Hanoi, Vietnam); Tung Vu (Posts and Telecommunications Institute of Technology, Vietnam); Long Pham and Duc-Tho Mai (Academy of Cryptography Techniques, Vietnam); Cong-Kha Pham (University of Electro-Communications (UEC), Japan)	Oral
1:45 PM	1571273620	Hardware Acceleration of Layer Normalization on SoC for Artificial Intelligence on the Edge	Ettore Napoli and Efstratios Zacharelos (University of Salerno, Italy)	Oral
2:00 PM	1571273719	A RISC-V Based Hardware Platform with CNN Acceleration for Real-Time Neonatal Seizure Detection	Lakshmi Iyer (Indian Institute of Technology, Bombay, India); Ankur Gupta, Shrey Aggarwal, Mithin K and Laxmeesha Somappa (Indian Institute of Technology Bombay, India)	Oral
2:15 PM	1571274113	High-Reliability O(1) Logic-Depth LDPC Decoder for Edge AI Storage	Yanling Jiang, Junyi Wang, Xihui Qian, Zhixiong Li, Yida Li and Longyang Lin (Southern University of Science and Technology, China)	Oral

1:15 PM– 2:30 PM	Session #16 Special Session: Specialized Hardware and Systems for Edge Large Language Model Acceleration and Application, Precision-Scalable and Reliability-Aware AI Circuits for Closed-Loop Edge Autonomy, Multimedia Systems and Applications for AI (4 papers) Venue: Diamond 3 Chairs: Duy-Hieu Bui, VNU, Hanoi			
Time	Paper #	Title	Authors	Type
1:15 PM	1571273554	A Low-Latency and Low-Hallucination Edge Compound AI System via RAG for Embodied Robotics	Mingde Duan, Yitong Shen, Zhenhao Wang, Yingjie Zhao, Yanwen Ding and Xuhui Du (Xidian University, China); Sai Zhang (Junhe Technology (Quzhou) Co., Ltd.); Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Oral
1:30 PM	1571273607	OPAL: Operator-Aware Precision-Sparsity Co-Scheduling for CIM-Based Transformer Inference	Yitong Shen, Can Liu and Yukun Fan (Xidian University, China); Jie Gan and Zhe Wang (China); Xin Jin (Beijing Smartchip Microelectronics Technology Company Limited, China); Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Oral
1:45 PM	1571274655	Stochastic-LLM: Exploring Sign-Magnitude Stochastic Computing for Low-Power Edge LLM Inference on FPGAs	Yanwen Ding (Xidian University, China); Shirui Zhao (HKUST, Hong Kong); Cheng Zhang, Mingde Duan, Yitong Shen, Xuhui Du, Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Oral
2:00 PM	1571276533	CIMBA: Hybrid State-Space Model Inference with Precision-Adaptive Hybrid Compute-in-Memory	Avadh Rajesh Harkishanka, Jintao Zhang, Yimin Wang and Xuanyao Fong (National University of Singapore, Singapore)	Oral
2:30 PM	Tea break			

3:00 PM– 4:00 PM	Session #17: Computer vision algorithms and architectures II (3 papers) Venue: Panorama 1 Chairs: Qing Zhang, Shenzhen University, China			
Time	Paper #	Title	Authors	Type
3:00 PM	1571267969	On the Accuracy Flip of Vision Transformers Under Ultra-Low-Bit Post-Training Quantization	Hyunha Hwang, Hyuk-Jae Lee and Xuan Truong Nguyen (Seoul National University, Korea (South))	Oral
3:15 PM	1571268473	LSTA-Net: A Spatiotemporally Decoupled Network with Single-Query Transformer for Low Power Event-Based Eye Tracking	Shangyu Yang, Zhijie Jian and Yanbo Lin (Tsinghua University, China); Hong Chen (Tsinghua, China)	Oral
3:30 PM	1571273530	ASIC Functional Cell-Level Classification on Infra-Red, in-Situ (IRIS) Data for Hardware Assurance	Kakuly Mittal (Nanyang Technological University, Singapore); Andrew Huang (Baochip, Inc., Singapore); Bah Hwee Gwee (NTU, Singapore)	Oral
3:00 PM – 4:00 PM	Session #18 Special Session: AI-Powered Electronic Design Automation for IC Design II (4 papers) Venue: Panorama 2 Chairs: Thanh-Loan Pham Nguyen, HUST, Vietnam			
Time	Paper #	Title	Authors	Type
3:15 PM	1571270399	AST-GWO: Adaptive Sigmoid Transfer Grey Wolf Optimizer for Improving Search Dynamics in Logic Synthesis	Dat Tan Tran, Thuc Duy Le, Gia Huy Duong and Hoang Trang (Ho Chi Minh City University of Technology, Vietnam)	Oral
3:30 PM	1571274086	Component-Centric Placement Using Deep Reinforcement Learning	Kart Leong Lim (Institute of Microelectronics, Singapore)	Oral
3:45 PM	1571274215	Multi-Objective Reinforcement-Learning-Based Fogging and Proximity-Aware Placement Method for Analog and Mixed-Signal Circuits	Mirvala Sadrafshari, Octavia A. Dobre and Lihong Zhang (Memorial University of Newfoundland, Canada)	Oral
4:00 PM	1571274985	GRPO-UVM: A Reinforcement Learning-Driven Verification Framework for AI Compute Systems	Jiaxiang Wang and Tingjie Yang (Xidian University, China); Shirui Zhao (HKUST, Hong Kong); Yingjie Zhao, Cheng Zhang, Zhenhao Wang, Jialong Wang, Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Oral

3:00 PM– 4:00 PM	Session #19 Special Session: Emerging Energy-Efficient Hardware and Computing Paradigms for Edge Intelligence II (4 papers) Venue: Diamond 2 Chairs: Xueqing Li, Tsinghua University, China			
Time	Paper #	Title	Authors	Type
3:15 PM	1571273984	Energy-Efficient SNN-Based Person Detection with Adaptive Burst Suppression	Alimatou Sadia Anike Memudu (Université Côte d'Azur, France & IMRA Europe, France); Jean Martinet (Université Côte d'Azur, France); Yuta Nakano (IMRA Europe, France)	Oral
3:30 PM	1571274209	A Novel Ferroelectric Computing-in-Memory Framework for Solving Edge Robotic Equations	Xueyang Bai, Bo Li, Xiguang Wu and Wei Mao (Xidian University, China); Guoyong Shi (Shanghai Jiao Tong University, China); Yan Liu and Genquan Han (Xidian University, China)	Oral
3:45 PM	1571274465	A Hardware-Efficient DWT-ConvLSTM Spatio-Temporal System for Arc Fault Early Detection at the Edge	Xuhui Du, Yingjie Zhao, Zhenhao Wang and Jialong Wang (Xidian University, China); Mingde Duan (Hangzhou Institute of Technology, Xidian University, China); Jie Gan and Zhe Wang (China); Xin Jin (Beijing Smartchip Microelectronics Technology Company Limited, China); Xiaoxi Li, Xiguang Wu, Bo Li, Yan Liu, Wei Mao and Genquan Han (Xidian University, China)	Oral
4:00 PM	1571275037	Hardware Implementation of Efficient Reconfigurable 2D FFT Processor for SAR Imaging	Wenbo Ding and Weiwei Shi, Bo Zhao (Shenzhen University, China)	Oral

3:00 PM– 4:00 PM	Grand Challenge SS'25 Venue: Diamond 3 Chairs: Wei Mao, Xidian University, China			
Time	Paper #	Title	Authors	Type
3:15 PM	1571268392	An Ultra-Efficient Inference Pipeline for Qwen2.5 on Yitian 710: Synergizing Fine-tuning, Quantization, and Inference Optimization	Wenyuan Zhao, Ruchao Fan, Zhao Cang, Yongming Tang and Hao Sun (Southeast University, China)	Oral
3:30 PM	1571270318	AICAS Grand Challenge 2025: Optimization and Application of Edge LLMs	Zhenhao Wang and Cheng Zhang, Tingjie Yang and Huanning Zhang (Xidian University, China); Guosheng Yu (T-HEAD Semiconductor Co., Ltd); Evens Pan (Arm Technology (China) Co., Ltd); Liang Chang (University of Electronic Science and Technology of China, China); Yuan Du and Li Du (Nanjing University, China); Zhezhi He and Yongfu Li (Shanghai Jiao Tong University, China); Bo Li and Wei Mao (Xidian University, China)	Oral
3:45 PM	1571271740	GRASP: A Gm/Id-Based Multi-Agent Reasoning Framework for Sizing Parameters in Amplifiers	Ji Zhuang, Zishuo Wang, Jinyi Shen, Zihao Chen and Fan Yang (Fudan University, China)	Oral
4:00 PM	1571273651	ELLME: A 280-Tokens/s Real-Time FPGA LLM Accelerator for End-to-End Qwen-2.5 Inference	Yujie Ma, Linqi Li, Yuman Zeng, Zhirui Huang and Chixiao Chen, Qi Liu and Haozhe Zhu (Fudan University, China)	Oral
4:00 PM	Tea break			

4:15 PM– 5:45 PM	Session #20: Emerging applications: IoT, healthcare, smart factories, environment, robotics, autonomous vehicles, drones, AR/VR, metaverse II (4 papers) Venue: Panorama 1 Chairs: Manh-Hiep Dao, VNU Hanoi			
Time	Paper #	Title	Authors	Type
4:15 PM	1571268423	Hardware-Aware Semantic Segmentation Engine for on-Chip Retinal Prosthesis Navigation	Arnav Agarwal (Indian Institute of Technology Bombay, India); Daksh Sawke (Indian Institute of Technology, Bombay, India); Laxmeesha Somappa (Indian Institute of Technology Bombay, India)	Oral
4:30 PM	1571268450	Enhancing Text Entry Efficiency in AR/VR with a Pinch- and Thumb-Click Gesture-Based Virtual Keyboard	Youngmock Cho, Donggyu Kim, Viduranga Munasinghe, Tae-Ho Lee and Hyuk-Jae Lee (Seoul National University, Korea (South))	Oral
4:45 PM	1571272160	QR-Coded Microstrip Antennas: End-To-End Co-Design of URL Extensions and RF Performance for IoT and IP Protection	Yuting Hou (Shanghai Jiao Tong University, China & Xi'an Railway Vocational Technical Institute, China); Chao Wang and Jinghua Zhang (Shanghai Jiao Tong University, China); Fu Qi Chua (Universiti Malaya, Malaysia); Liyuan Du (Xi'an Railway Vocational Technical Institute, China); Mayank Kumar Singh (Shanghai Jiao Tong University, China); Guoliang Feng (Xi'an Railway Vocational Technical Institute, China); Yongfu Li (Shanghai Jiao Tong University, China)	Oral
5:00 PM	1571273958	ULASC: An Ultra-Lightweight Acoustic Scene Classification Model for IoT Devices	Xiben Jiao, Yuanchang Sun, Xiaotian Wei, Fan He, Jiacheng Liang, Rui Hong, Haipeng Liu and Xingdan Cai (University of Electronic Science and Technology of China, China); Weixian Kang (Chip-Artisan Technology (Chengdu) Co. Ltd, China); Xinyi Jing, Jiahao Liu and Jun Zhou (University of Electronic Science and Technology of China, China)	Oral

4:15 PM – 5:45 PM	Session #21: Hardware/Software Co-Design and Design Automation for AI systems II (5 papers) Venue: Panorama 2 Chairs: Zhen Zhuang, The Chinese University of Hong Kong			
Time	Paper #	Title	Authors	Type
4:15 PM	1571268164	Front-End Quantization Can Recover Accuracy Under Bandwidth-Induced Covariate Shift in Keyword Spotting	Nazareno Sacchi (ETHZ - ETH Zurich, Switzerland & CSEM, Switzerland); Taekwang Jang (ETHZ, Switzerland); Komail Badami (CSEM, Switzerland)	Oral
4:30 PM	1571268199	Energy Efficient Analog Neural Network via SHAPE: Sensitivity-Aware Heterogeneous Power Allocation Framework	Leyang Jiang (Shanghai Jiaotong University, China); Siyun Ding, Ruge Xu, Xinfei Guo and Xuyang Lu (Shanghai Jiao Tong University, China)	Oral
4:45 PM	1571268380	Performance-Aware Rank Adjustment for Hardware-Friendly Fine-Tuning in Edge AI Accelerators Exploiting Sparsity and Scalability	Bo-Yang Lin and Cheng-Han Ho (National Central University, Taiwan); Pei-Yun Tsai (National Taiwan University, Taiwan)	Oral
5:00 PM	1571268438	LPAC-MoE: Lossless and Prefetch-Aware FP8 MoE Compression via Algorithm-Hardware Co-Design	Qiye Ding, Dingyang Zou and Zhongfeng Wang (Nanjing University, China)	Oral
5:15 PM	1571268065	TRACE: Temporal Reuse and Communication-Aware Mapping for Heterogeneous CIM Chiplet Systems	Zixuan Wang, Abhishek Tyagi and Xuanyao Fong (National University of Singapore, Singapore)	Oral

4:15 PM – 5:45 PM	Session #22 Special Session: Compute-in-Memory Co-Design: From Emerging Devices to AI Application Optimization (5 papers) Venue: Diamond 2 Chairs: Quoc-Hung Nguyen, VNU, Hanoi			
Time	Paper #	Title	Authors	Type
4:15 PM	1571273940	OpenMENA: An Open-Source Memristor Interfacing and Compute Board for Neuromorphic Edge-AI Applications	Ali Safa (Hamad Bin Khalifa University, Qatar); Farida Ali Mohsen (College of Science and Engineering, Qatar); Zainab Ali, Bo Wang and Amin Bermak (Hamad Bin Khalifa University, Qatar)	Oral
4:30 PM	1571274720	A CIM-Optimized Navigation Algorithm Based on Grid Cells for Edge Applications	Mu Rongjun (Zhejiang University, China); Jiabao Ye (China); Bing Chen (Xidian University, China); Yong Ding (Zhejiang University, China)	Oral
4:45 PM	1571274728	Phase-Domain Neural Computation Based on Multi-Injection Locked Oscillators	Badets Franck (CEA, France); Kamal Danouchi (CEA-Leti, France); Jean-Baptiste Casanova (CEA-List, France)	Oral
5:15 PM	1571274868	Beyond Evolutionary Search: A General Reinforcement Learning Framework for Scalable Hardware Trojan Mitigation on Evolvable Hardware Platforms	Gaurav Ravindra Nayak and Niranjana Gopal and Madhav Rao (International Institute of Information Technology, Bangalore, India)	Oral
5:30 PM	1571274944	Modeling of Thin-Film Transistors and Memristors from Fabricated Devices with Applications in Memristor-Based Circuit Design for Neuromorphic AI Computing	Ngoc-Lam Pham Le (Vietnam National University – Hanoi); Phu-Quan Pham (Vietnam National University-HCM); Viet-An Dang, Hoang-Viet Vu, Thuat Nguyen-Tran and The-Vinh Tran (Vietnam National University – Hanoi)	Oral
4:15 PM– 5:45 PM	Grand Challenge SS'26 Venue: Diamond 3 Chairs: Wei Mao, Xidian University, China			
Time	Paper #	Title	Authors	Type
4:15 PM	1571332291	VersaVLM: A Phase-Specialized FPGA NPU with Log8 Attention for VLM Inference	Zhenyang Gu, Jiakun Shen, Xichen Wang, Lanqi Ma and Gengsheng Chen (Fudan University, China)	Oral
4:30 PM	1571334356	Overcoming the Kernel Launch Bottleneck: 11x Faster VLM Inference with Operator Fusion and CUDA Graphs	Thomas Yeung (TWGHs Kap Yan Directors' College, Hong Kong); YiLin Zhu (East China Normal University, China)	Oral
4:45 PM	1571335416	AICAS Grand Challenge 2026: Efficient VLM Inference on AI Chips and FPGA-Based Edge Systems	Cai Binglun (Shanghai Jiao Tong University, China); Linqi Li (Fudan University, China); Kang You (Shanghai Jiao Tong University, China); Bo Li and Wei Mao (Xidian University, China); Liang Chang (University of Electronic Science and Technology of China, China); Yuhang Zhang	Oral

			(East China Normal University, China); Yiming Gan (Institute of Computing Technology, Chinese Academy of Sciences, China); Shimeng Qian (Pingtounge Semiconductor Co Ltd, China); Yuan Du and Li Du (Nanjing University, China); Yongfu Li (Shanghai Jiao Tong University, China); Haozhe Zhu (Fudan University, China); Zhezhi He (Shanghai Jiao Tong University, China)	
5:00 PM	1571335654	UniVLM: Unified ViT-LLM Accelerator for Efficient End-to-End VLM Inference on FPGA	Le Zhong, Linfeng Zhong, Chunbo Li, Huifeng Wen, Yunpeng Wang and Haoyang Ping (Peking University, China); Meng Li (Peking University, China); Yuan Wang (Peking University, China)	Oral
5:15 PM	1571335704	End-to-End System Optimization for Low-Latency Single-Request Vision Language Model Inference	Damhyun Lee, Jongwon Lee, Yujun Heo, Sihwa Lee and Jungwook Choi (Hanyang University, Korea (South))	Oral
5:30 PM	1571335705	Memory-Centric VLM Deployment for Bandwidth-Constrained Heterogeneous Systems	Yuqi Guo (University of Electronic Science and Technology of China, China); Ruiqi Tang (Fudan University, China); Chuqi Zhao (Vanderbilt University, USA); Yucheng Sun (University of Electronic Science and Technology of China, China); Bo Yan (University of Electronic Science and Technology of China)	Oral
5:45 PM	1571335713	CASA: Context-Anchored Speculative Attention and System Co-Design for High-Throughput Vision-Language Inference	Yongkang Huang, Tianjiao Yang, Shuyan Chen and Jiang Zhu (China)	Oral
7:00 PM	Banquet & Award Ceremony Shuttle buses will depart at the conference venue (the hotel's lobby) from 18:00 to the restaurant.			

Friday, September 18, 2026

Day 3: Friday, 18 September 2026 Venue: Wyndham Legend Halong Hotel Address: 12 Ha Long Street, Bai Chay Ward, Quang Ninh Province 02033				
9:00 AM – 10:30AM	Session #23: Neuromorphic and bioinspired circuits, processors, systems, and their applications I (5 papers) Venue: Panorama 1 Chairs: Manh-Hiep Dao, VNU, Hanoi			
Time	Paper #	Title	Authors	Type
9:00 AM	1571262851	BCPNN Torch: A PyTorch Simulation Framework for BCPNN-Based Associative Memory	Xiaoze Yan, Deyu Wang, Yu Zhuoquan and Huidong Ji (Fudan University, China); Yu Yang (KTH Royal Institute of Technology, Sweden); Qi Jiang (Fudan University, China); Ahmed Hemani (Royal Institute of Technology, Sweden); Anders Lansner (Royal Institute of Technology (KTH), Sweden); Jiawei Xu (University of Macau, Macao); Lirong Zheng (Fudan University, China); Zhuo Zou (Fudan University & KTH Royal Institute of Technology, Sweden)	Oral
9:15 AM	1571267548	Adversarial Vulnerability of Neuromorphic Processors Demonstrated on Loihi Chip	Spyridon Raptis and Haralampos-G. Stratigopoulos (Sorbonne Université, CNRS, LIP6, France)	Oral
9:30 AM	1571267931	A Pulse Width Modulation-Based Compensation Circuit for Improving Linearity of Memristor Conductance Modulation	Kyusik Mun, Hyeongcheol Jo, Jaehyun Kang and Donghyuk Shin (Korea Institute of Science and Technology, Korea (South)); Joon Young Kwak (Ewha Womans University, Korea (South)); Jongkil Park, Suyoun Lee, Inho Kim, Jong-Keuk Park, Changhyuk Lee and Seongsik Park (Korea Institute of Science and Technology, Korea (South)); Hyung-Min Lee (Korea University, Korea (South)); Yeonjoo Jeong and Jaewook Kim (Korea Institute of Science and Technology, Korea (South))	Oral
9:45 AM	1571267980	A Continuous-Time QUBO Solver Using a Chain-RRAM-Based Hopfield Neural Network	Congcong Hong and Zhong Sun (Peking University, China)	Oral
10:00 AM	1571268029	Hyperdimensional Computing Training and Inference with Eight-Bit Floating-Point Formats	Andrea Marcelli and Marco Angioli (Sapienza University of Rome, Italy); Denis Kleyko (Örebro University & RISE Research Institutes of Sweden, Sweden); Marco Pisani (Sapienza University of Rome, Italy); Amy Loutfi	Oral

			(Örebro University, Sweden); Mauro Olivieri (Sapienza University of Rome, Italy)	
9:00 AM – 10:30AM	Session #24: In-Memory Computing II (6 papers) Venue: Panorama 2 Chairs: Anh Tuan Do, A*STAR, Singapore			
Time	Paper #	Title	Authors	Type
9:00 AM	1571268332	RINAC: An Energy-Efficient Computing-in-Memory SNN Accelerator Using Redundant Input-Aware Approximate Computing	Wei-Chi Huang and Nathaniel Rensly (National Tsing Hua University, Taiwan); Meng-Fan Chang (National Tsing Hua University (NTHU), Taiwan); Kea Tiong Tang (National Tsinghua University, Taiwan)	Oral
9:15 AM	1571268483	Robust 3-Bit/Cell ReRAM CiM Against Device Non-Idealities by Discrete Shift RTN Suppression and Max Sigma Gaussian Optimized QAVAT	Elliot Yushi Abarra and Naoko Misawa (Tokyo University, Japan); Chihiro Matsui and Ken Takeuchi (The University of Tokyo, Japan)	Oral
9:30 AM	1571268509	MISS-CiM: Mixed-Signal SRAM Based Compute-in-Memory Systolic Accelerator for Efficient DNN Inference in 2 nm GAA CMOS	Sai Shubham (Indian Institute of Technology, Bombay, India); Harsh Kakadiya, Karan Jaiswal, Megha S and Veeresh Deshpande (Indian Institute of Technology Bombay, India)	Oral
9:45 AM	1571273376	An Energy-Efficient, SCA-Resilient AES Encryption Accelerator Design with Dual-Rail SRAM LIM Macro for Edge Devices	Zhongyu He and Zixuan Shen (Huazhong University of Science and Technology, China); Nay Aung Kyaw (Nanyang Technological University, Singapore); Yuansheng Zhao (Huazhong University of Science and Technology, China); Shirui Sheng (Nanyang Technological University, Singapore); Kwen Siong Chong (Singapore); Bah Hwee Gwee (NTU, Singapore); Chao Wang (Huazhong University of Science and Technology, China)	Oral
10:00 AM	1571273548	Bit Error Tolerance Analysis of High Inference Accuracy AI Using Low-Voltage Floating-Point SRAM-CiM	Tetsuro Mizuno, Seiji Adachi, Yoshinari Wada, Ayane Matsuzaki, Kota Ando, Tetsuya Asai and Takao Marukame (Hokkaido University, Japan)	Oral
10:15 AM	1571274583	A Sparsity-Aware Adaptive Column Replication Framework for Reliable and Energy Efficient RRAM CiM Design	Hyeon-gwon Kim, Seung-Eon Hwang, Dongwoo Lew and Jongsun Park (Korea University, Korea (South))	Oral
10:30 AM	Tea Break (15 minutes)			

10:45AM – 11:45AM	Session #25: Neuromorphic and bioinspired circuits, processors, systems, and their applications II (6 papers) Venue: Panorama 1 Chairs: Duc-Tho Mai, ACT, Vietnam			
Time	Paper #	Title	Authors	Type
10:45 AM	1571268340	A Spiking Neuromorphic Architecture for Belief Propagation Decoding of LDPC Codes	Shiang-Yi Chi (National Yang Ming Chiao Tung University, Taiwan); Li-wei Liu and Hsie-Chia Chang (National Chiao Tung University, Taiwan)	Oral
11:00 AM	1571268354	A 1k-Neuron 16bit Edge-FPGA-SoC Reservoir Computing Processor Exceeding 1Mps Input Rate by Zero Skipping for Variable Unstructured Sparsity	Takashi Oshima (Hitachi Ltd., Japan); Yuya Takanuki and Kunihiro Fujisaki (Hitachi Information & Telecommunication Engineering, Ltd., Japan)	Oral
11:15 AM	1571268386	A Firework Inspired Routing Scheme with Single-Packet in-Network Phase Transition for Neuromorphic Systems	Qi Jiang, Fanxi Yang, Lufei Fan, Yi Zheng, Yuhan He, Xun He, Xiaozhe Yan and Xirui Peng (Fudan University, China); Hannu Tenhunen (KTH Royal Institute of Technology, Sweden & Ingenium Machina Ltd., Finland); Lirong Zheng (Fudan University, China); Zhuo Zou (Fudan University & KTH Royal Institute of Technology, Sweden)	Oral
11:30 AM	1571268466	PSA: A Pruning-Friendly Sparse-Aware Accelerator for Efficient Sparse Convolution in SNNs	Sanhe Fu and Kainan Wang (Zhejiang University, China); Bo Wang (Singapore University of Technology and Design, Singapore); Aili Wang (Zhejiang University, China)	Oral
11:45 AM	1571273646	RHYME: A Resource-Efficient Hybrid Neuromorphic Hardware System for EMG Gesture Classification	Yuvraj Chowhan (Indian Institute of Technology (IIT) Delhi, India); Chithambara Moorthii J and Anmol Singla (Indian Institute of Technology Delhi, India); Manan Suri (Indian Institute of Technology - Delhi, India)	Oral
12:00 PM	1571273836	L-SPINE: A Low-Precision SIMD Spiking Neural Compute Engine for Resource-Efficient Edge Inference	Sonu Kumar and Mukul Lokhande (Indian Institute of Technology Indore, India); Santosh Kumar Vishvakarma (IIT, Indore, India)	Oral

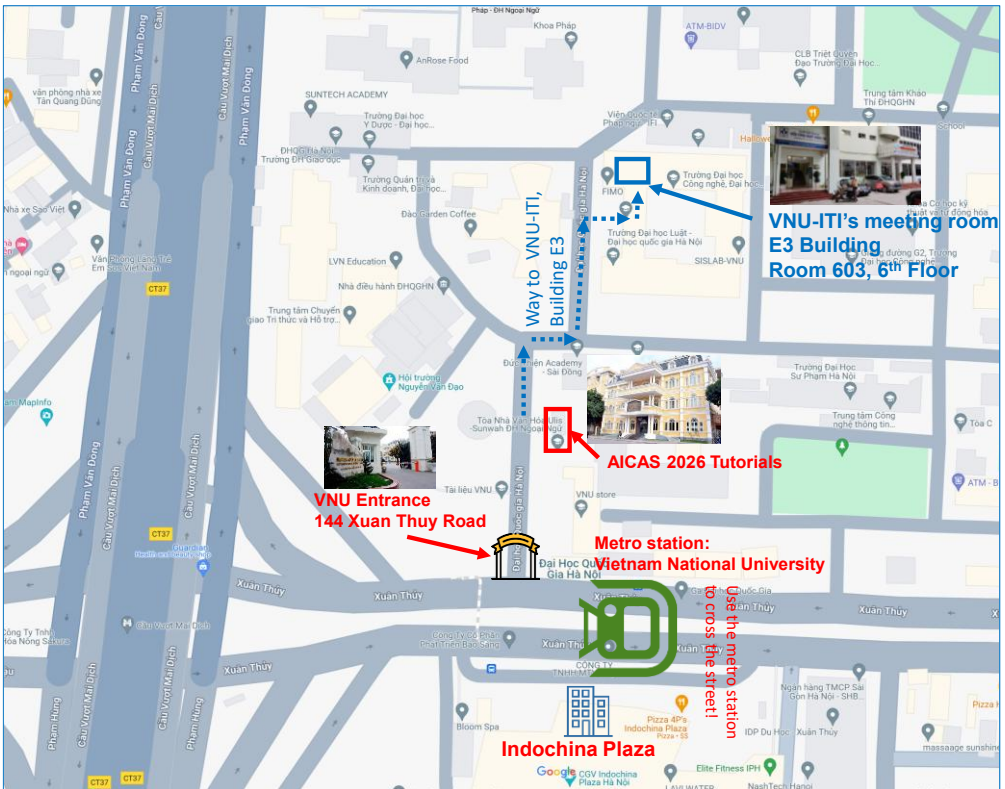
10:45AM – 11:45AM	Session #26 Special Session: Compute-in-Memory Co-Design: From Emerging Devices to AI Application Optimization II (4 papers) Venue: Panorama 2 Chairs: Qing Zhang, Shenzhen University, China			
Time	Paper #	Title	Authors	Type
10:45 AM	1571274967	Explainable and Robust Fake-News Detection System via LLM-Based Credibility and Context Analyses	Tzu-Shuo Peng, Oscar Tzyh-Chiang Chen and Yu Hsiang Hung (National Chung Cheng University, Taiwan); Manh-Hung Ha (Vietnam National University, Vietnam)	Oral
11:00 AM	1571275092	A Hardware-Aware Optimized YOLOv8 for Real-Time Crack Detection at the Edge	Jingjun Sun, Tan Peng, Jiawen Chen, Bashir M. Al-Hashimi and Kai Xu (Kings College London, United Kingdom (Great Britain))	Oral
11:15 AM	1571275722	Process Variation in Probabilistic Compute-in-Memory: Challenge and Opportunity	Steven Davis, Zephan M. Enciso, Likai Pei and Ningyuan Cao (University of Notre Dame, USA)	Oral
11:30 AM	1571275830	CIMERA: Compute-In-Interconnect and Memory with Reconfigurable Precision for LLM Inference	Yue Jiet Chong, Yimin Wang, Wei Zhang and Xuanyao Fong (National University of Singapore, Singapore)	Oral
12:15 PM	Lunch			
1:30 PM – 5:15 PM	Workshop #1: The 2nd IEEE Chiplet Workshop on Circuits and Systems for Next-Generation AI Computing Venue: Panorama 1 Chairs: Chixiao Chen, Fudan University, China; Anh Tuan Do, A*STAR, Singapore.			
Time	Paper #	Title	Authors	Type
1:30 PM		Opening Remarks		
1:40 PM	CSNAIC-1	The Carbon cost of Specialization: Rethinking AI Hardware in the Chiplet Era	Tulika Mitra, National University of Singapore	Oral
2:20 PM	CSNAIC-2	Scalable 3D Compute-in/near-Memory Systems with Reusable Active Interposer	Chixiao Chen, Fudan University	Oral
3:00 PM	CSNAIC-3	Low-Power Wireline/Optical Transceivers for Next-Generation AI Applications	Quan Pan, Southern University of Science and Technology	Oral
3:40 PM	Tea Break (15 minutes)			
3:55 PM	CSNAIC-4	Progress on Crosstalk Cancellation Techniques for High-Speed Parallel Interconnects	Yuan Du, Nanjing University	Oral

4:35 PM	CSNAIC-5	A 16Gb/s/pin 0.51pJ/bit Single-Ended NRZ Transceiver with VDDQ Ripple Compensation for Memory Interfaces	Wenning Jiang, Fudan University	Oral
1:30 PM – 2:00 PM	Industry talk: HiFloat Data Format Principles, Practices and Standards Haonan Zhang, Huawei, China Venue: Panorama 2 Chairs: Manh-Hiep Dao, VNU, Hanoi			
2:00 PM – 5:15 PM	Workshop #2: Smart Sensor Circuit and Intelligent Systems for Emerging Applications Venue: Panorama 2 Chairs: Qing Zhang, Shenzhen University, China; Milin Zhang, Tsinghua University, China			
Time	Paper #	Title	Authors	Type
2:00 PM		Opening Remarks		
2:10 PM	S2CAS-1	Neuromorphic Smart Nose-on-a-Chip with On-Chip Learning for Pulmonary Diagnosis	Hong Chen, Tsinghua University, China	Oral
2:55 PM	S2CAS-2	Towards Trustworthy In-Vehicle AI Systems: Dynamic Watermarks for Real-Time Intrusion Detection	Yi (Estelle) Wang, AUMOVIO, Singapore	Oral
3:40 PM	Tea Break			
3:55 PM	S2CAS-3	Ultra-Low-Power and High Accuracy AI Voice Control Solution: Co-Optimization of Algorithm, Hardware and Toolchain for Edge AI Devices	Jianbiao Xiao, Chip-Artisan Tech. Corp., China	Oral
4:40 PM	S2CAS-4	Approximate Computing Circuit Techniques in Efficient Smart Edge Systems - from Biomedical/Vision Recognition to SAR Imaging	Weiwei Shi, Shenzhen University, China	Oral

Information

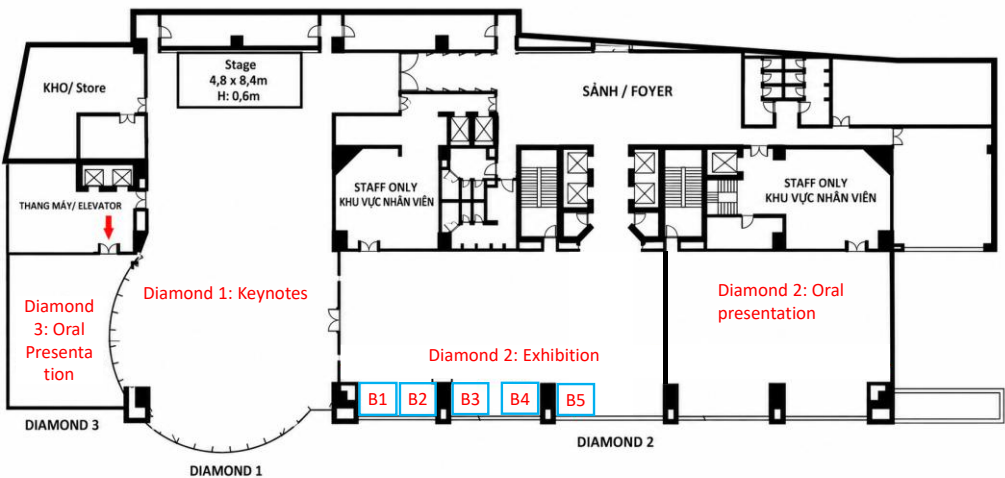
Tutorial Venue Map

Please note that participants should take Grab (Android/iOS apps) or Green SM to the venue. If you take a traditional taxi, please ask the driver to drop you at VNU Entrance (144 Xuan Thuy Road) or the Indochina Plaza Hanoi (IPH) building (opposite the VNU main entrance). **You can use the metro station to cross the street safely. The metro station has lifts.**

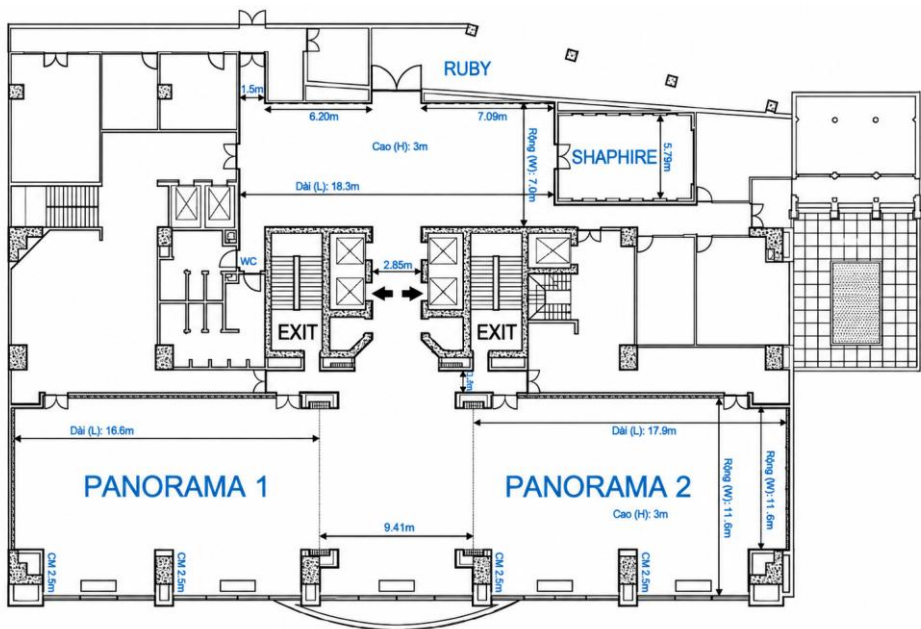


Conference Venue Map

Second floor



Third floor





URL: <https://2026.ieee-aicas.org/publication/>

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Duy-Hieu Bui

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Welcome reception

Time: 19:30, September 16, 2026 *(Shuttle bus is available at 18:00)*
Location: Dragon Pearl Cave (45 minutes from Ha Long, Vietnam)
Address: Diem Thuy Area, Cam Pha, Quang Ninh, Vietnam
Shuttle buses will depart from the conference venue (the hotel's lobby) at 18:00 to the restaurant.

Banquet & Award Ceremony

Time: 19:00, September 17, 2026 *(Shuttle bus is available at 18:00)*
Location: Paradise Delight
Address: Tuan Chau International Marina, Ha Long Bay, Vietnam
Shuttle buses will depart from the conference venue (the hotel's lobby) at 18:00 to the restaurant.

Wi-Fi Access

Tutorials at Xuan Thuy campus

Venue	Jonathan KS Choi Center (Sunwah building)	
Access point	SUNWAH	ULISWIFI-MeetingRoom
Password	1955@ulis	dhnn1955

Conference venue in Ha Long Bay

Venue	Wyndham Legend Ha Long	
Access point	AICAS 2026	AICAS 2026
Password	88888888	aicas2026

Transportation

Transportation apps

An easy way to book taxis or motorbikes is through ride-hailing apps such as Grab and Green SM. Grab and Green SM are Vietnam's Uber and can be used to book a car, taxi, or motorbike.

Here is a guide to using ride-hailing apps for transportation.

Step 1: Open the app by scanning the QR code below.



Grab

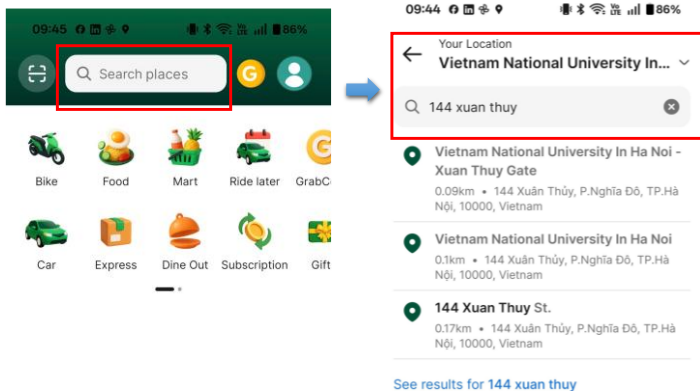


Green SM

Step 2: Sign up or log in to your account

- Initiate by selecting *Continue*
- Enter the phone number in use → *Continue*, and enter the OTP code sent by Taxi GREEN SM via SMS.
- Enter Account Information, including Full Name and Password → *Sign Up*.
- After that, Verify Phone Number → *Continue* → enter Password → *Login*.

Step 3: Enter your pickup and destination locations.



Step 4: Choose your vehicle type and confirm your booking. You can pay by cash/credit card.

The Information Technology Institute, Vietnam National University, Hanoi

Vietnam National University, Hanoi (VNU) is one of Vietnam's two national universities and the country's largest comprehensive higher education and research center. Its history dates back to the University of Indochina, established in 1906, followed by Vietnam National University in 1945 and the University of Hanoi in 1956. VNU was reorganized in 1993 by integrating the University of Hanoi and other leading institutions. Today, VNU plays a key role in education, research, innovation, and developing highly qualified human resources for Vietnam.

Established in 2001, the Institute of Information Technology (VNU-ITI) is a member institute of VNU. ITI provides doctoral and master's education in Information Technology, Computer Science, and Engineering, while conducting scientific research, technology development, consulting, and technology transfer. The Institute also offers advanced training in emerging areas such as Artificial Intelligence (AI), Data Science, Blockchain, and Digital Transformation.

VNU-ITI currently comprises seven research centers in strategic and emerging technologies: (1) Information Safety and Security; (2) Applied Artificial Intelligence; (3) Integrated Circuits and Applications; (4) Semiconductor and Advanced Materials; (5) Quantum Technologies; (6) AI in Sustainable Development; and (7) UAV and Robotics.

Building on these centers, ITI conducts multidisciplinary research in AI and Data Science, VR/AR, Image and Video Processing, Blockchain, Internet of Things (IoT), Embedded Systems, Integrated Circuit (IC) and FPGA Design, Semiconductor Technologies, Quantum Technologies, UAVs, Robotics, and related fields.

VNU-ITI maintains an extensive international network with universities, research institutes, technology organizations, and enterprises in the United States, China, France, Germany, Italy, Japan, Korea, Singapore, Malaysia, and other countries. These partnerships support joint research, international education and training programs, technology development, and researcher exchange. ITI researchers regularly publish in WoS- and Scopus-indexed journals and conferences, with increasing emphasis on intellectual property, technology transfer, and research-driven innovation.

Through its multidisciplinary research centers and international partnerships, VNU-ITI aims to strengthen its position as a leading research and innovation hub in ICT, semiconductors, AI, and emerging technologies, contributing to VNU's internationalization and Vietnam's digital and technological transformation.

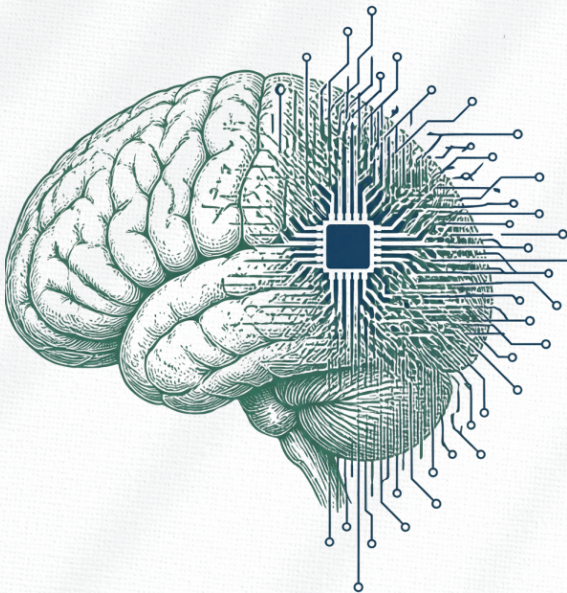
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